Digital Egypt Pioneers

FORTINET CYBERSECURITY ENGINEERING TRACK

FINAL PROJECT

IMPLEMENTING VPN SOLUTIONS WITH FORTIGATE

Submitted By:

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Submitted To:

- **Dr. Hussein Harb**

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Introduction

Virtual Private Networks (VPNs) are fundamental technologies that enable secure communication over untrusted networks. As modern organizations increasingly depend on distributed workforces, cloud services, and multi-site infrastructures, VPNs provide confidentiality, integrity, and reliable remote access to corporate resources. They protect traffic using encryption and tunneling mechanisms, allowing users and branches to exchange data securely as if they were on the same private network.

Fortinet's FortiGate platform offers a robust suite of VPN solutions that combine security, performance, and ease of deployment. Beyond basic connectivity, FortiGate integrates advanced capabilities such as threat inspection, identity-based access, and intelligent routing through SD-WAN. These features make it a powerful option for enterprises requiring both secure remote access and scalable site-to-site connectivity.

This project explores three major VPN implementations using FortiGate. In the first stage, an SSL VPN is configured to enable secure remote user access. The second stage establishes an IPsec site-to-site tunnel between two FortiGate devices to support branch interconnectivity. Finally, the project integrates FortiGate SD-WAN to optimize VPN traffic by selecting the best available WAN link for performance and reliability. The results of these implementations are documented, tested, and analyzed to demonstrate how Fortinet technologies can deliver efficient and secure VPN solutions.

Project Objectives

The main objective of this project is to design, configure, and evaluate secure VPN solutions using FortiGate. The project aims to explore how different VPN technologies address real-world requirements, such as secure remote access, site-to-site communication, and traffic optimization across multiple WAN links.

To achieve this, the project focuses on three key implementation stages:

1. **Understanding VPN Concepts and SSL VPN Setup**
   Study different types of VPNs and their use cases, then configure an SSL VPN on FortiGate to enable secure remote user access.

2. **Implementing IPSec VPN Connectivity**
   Establish a secure IPsec tunnel between two FortiGate devices, enabling encrypted site-to-site communication between private networks.

3. **Integrating SD-WAN with VPN Solutions**
   Configure SD-WAN features on FortiGate to monitor WAN links and intelligently route VPN traffic to improve reliability and performance.

Through these tasks, the project aims to demonstrate practical VPN deployment steps, compare different VPN methods, and illustrate how SD-WAN enhances VPN traffic stability in enterprise environments.

Theoretical Background

1. Virtual Private Networks (VPNs)

A **Virtual Private Network (VPN)** is a secure communication method that creates an encrypted tunnel between private or public networks. VPNs allow confidential data to travel across untrusted infrastructure—such as the internet—while maintaining confidentiality, integrity, and authentication.

Key Security Principles

- **Confidentiality** – Preventing unauthorized access to transmitted data through encryption.
- **Integrity** – Ensuring data is not modified during transit.
- **Authentication** – Verifying the identity of endpoints or users.

VPNs are widely used in enterprise environments to:

- Enable remote employee access to internal resources
- Connect branch offices securely (site-to-site)
- Improve cloud, IoT, or data center connectivity

2. SSL VPN

Concept

SSL VPN (Secure Sockets Layer Virtual Private Network) uses HTTPS (TLS) to secure communication between a user and the organization's internal resources. Unlike IPsec tunneling, SSL VPN works at the application layer (Layer 7) and is delivered through a web browser or VPN client.

Characteristics

- **Remote-access focused** — best for employees connecting from outside the network
- **Port-friendly** — runs over TCP/443, bypassing network restrictions

- **User authentication** — credentials, certificates, or MFA
- **Client-based or clientless** — depending on the deployment

Use Cases

- Secure employees accessing internal portals
- Remote management of servers or applications
- BYOD scenarios (phones, tablets, laptops)

In this project, the SSL VPN provides easy remote access to network resources through FortiGate, forming the foundation for VPN learning and validation.

3. IPsec VPN

Concept

IPsec (Internet Protocol Security) is a framework of protocols that secures IP communication at the network layer (Layer 3). It is the most common standard for site-to-site or branch-to-branch encryption.

Core Components

1. **IKE (Internet Key Exchange)** – Negotiates cryptographic parameters and authentication.
2. **ESP (Encapsulating Security Payload)** – Encrypts and protects payload data.
3. **Tunnel Mode** – Wraps original IP packets inside encrypted IPsec packets.

Advantages

- Strong encryption (AES, 3DES)
- High throughput for branch networks
- Interoperability between vendors
- Suitable for static or permanent tunnels

Use Cases

- Linking corporate headquarters to branch offices
- Secure inter-data center communication
- IoT/M2M connections

In this project, an IPsec tunnel is established between two FortiGate devices, simulating branch connectivity and validating tunnel integrity through connectivity testing.

4. SD-WAN (Software-Defined Wide Area Network)

Concept

SD-WAN is a network management approach that dynamically monitors WAN link performance and routes traffic based on real-time conditions. It allows enterprises to use multiple ISP links simultaneously and apply intelligent routing rules to maximize performance.

How SD-WAN Enhances VPNs

- Monitors latency, jitter, and packet loss
- Automatically selects the optimal WAN path
- Distributes overlays (e.g., IPsec) across multiple WAN links
- Provides failover during link degradation or failure

Key Benefits

- Improved end-to-end VPN performance
- Application-aware routing (VoIP, streaming, corporate apps)
- Higher availability through link redundancy
- Reduced operational cost vs MPLS

In this project, SD-WAN is configured on FortiGate to optimize VPN traffic, demonstrating how modern enterprises improve service quality without compromising security.

Implementation and Results

FortiGate SSL VPN Configuration

1. Network Design (Week 1)

The lab environment for Week 1 uses the following setup:

Laptop → Home Router → VMware → FortiGate VM

- **WAN (port1):** Receives DHCP IP from home router
- **LAN (port2):** Static IP 192.168.100.1/24
- **SSL-VPN Address Pool:** 10.212.134.200 – 10.212.134.210
- **Access:** FortiGate Web GUI

2. Week 1 Scope

The following tasks are included in Week 1:

- Configure WAN and LAN interfaces
- Create Local User for VPN authentication
- Create User Group for SSL VPN users
- Configure SSL-VPN Settings
- Configure SSL-VPN Portal (Tunnel Mode)
- Assign Tunnel Address Pool
- Create Firewall Policies for VPN traffic
- Document all steps with screenshots

3. Summary

Week 1 focuses on understanding VPN concepts and implementing the complete SSL VPN remote access configuration on FortiGate.
All components (interfaces, users, portal, address pool, and policies) are prepared for testing in the upcoming project weeks.

4. Configuration Steps

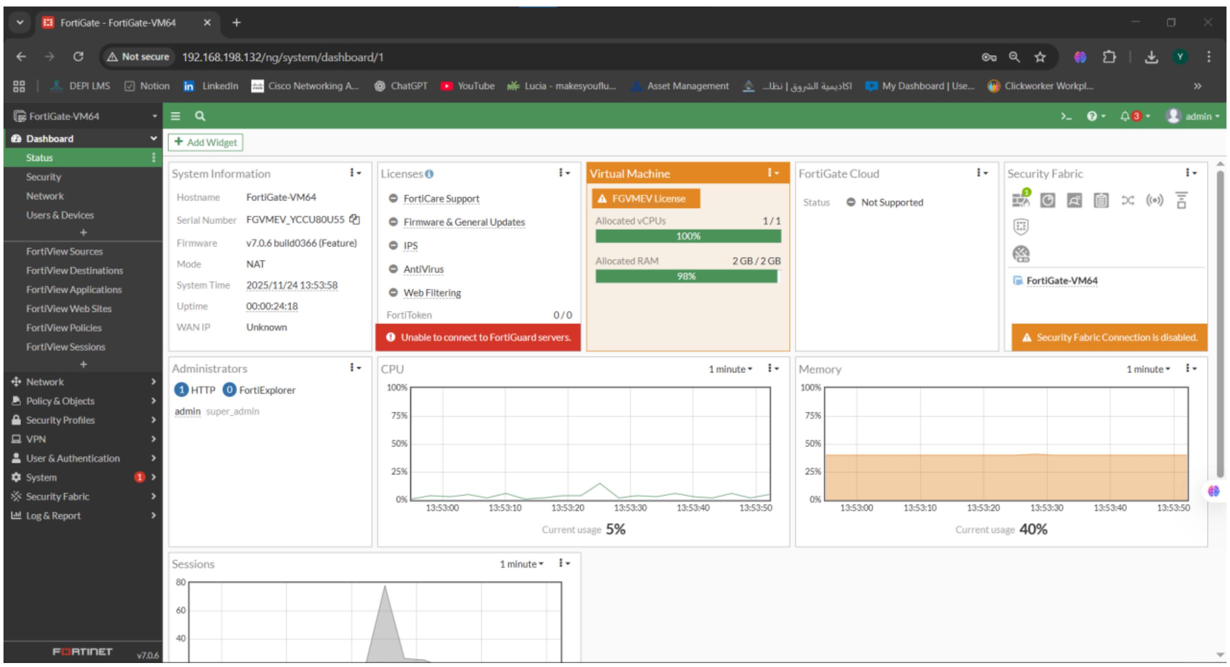


Figure 1 - Shows the main system summary, firmware version, uptime, and status of all interfaces.

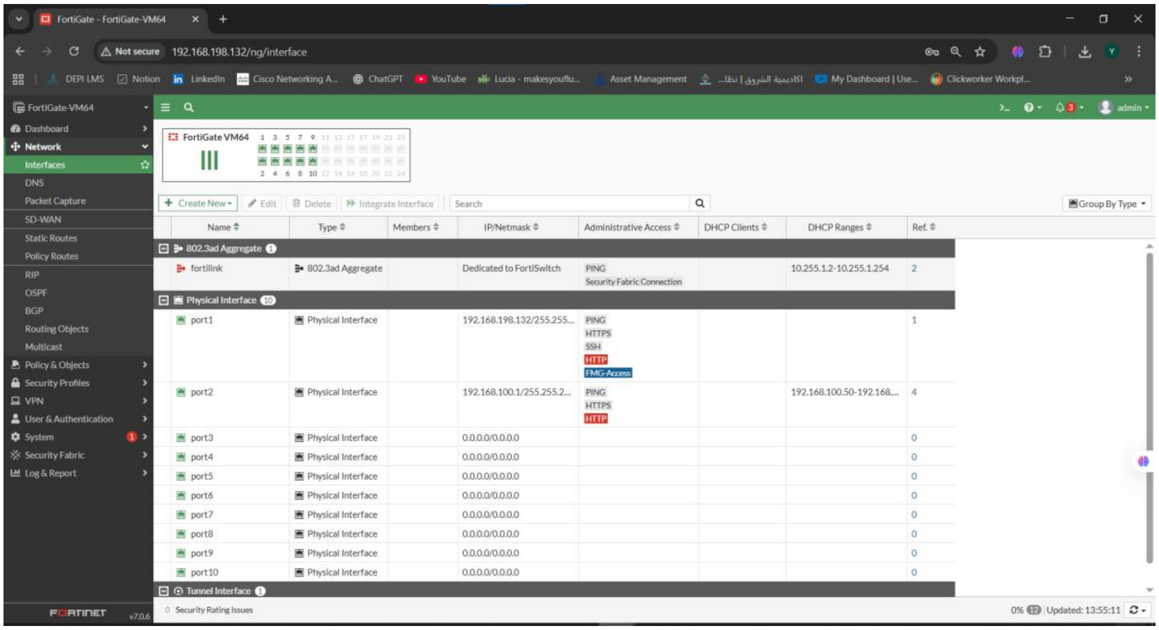


Figure 2 –(Network Interfaces Overview) Displays all FortiGate interfaces including port1 (WAN) and port2 (LAN).

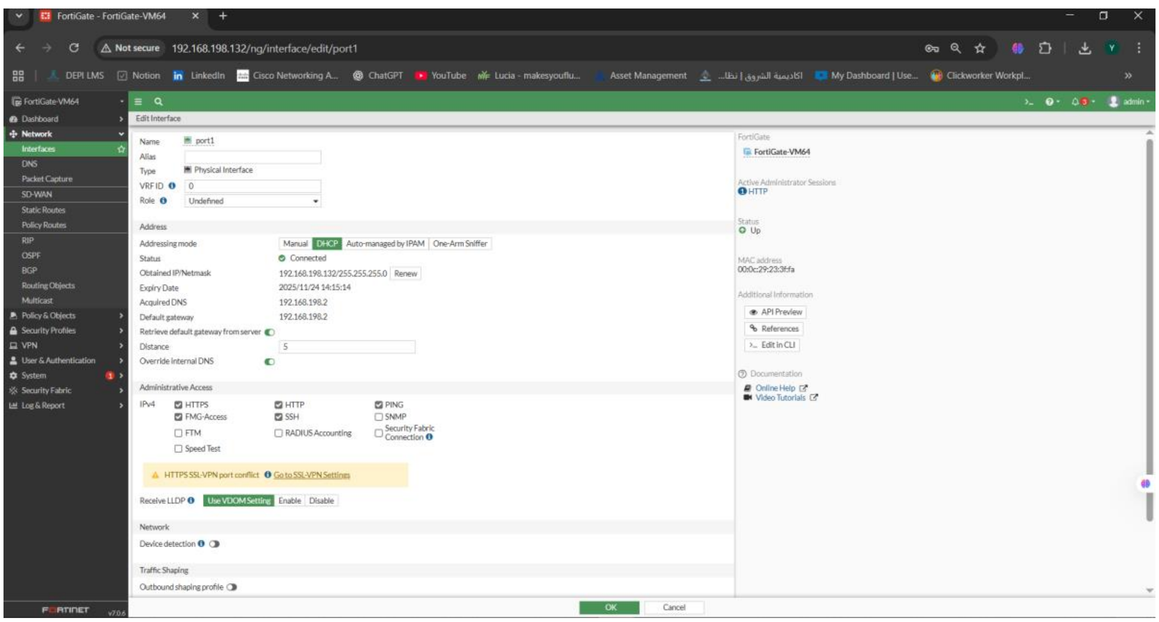


Figure 3 – (Port1 (WAN) Configuration) Port1 configured as the WAN interface using DHCP with assigned IP address.

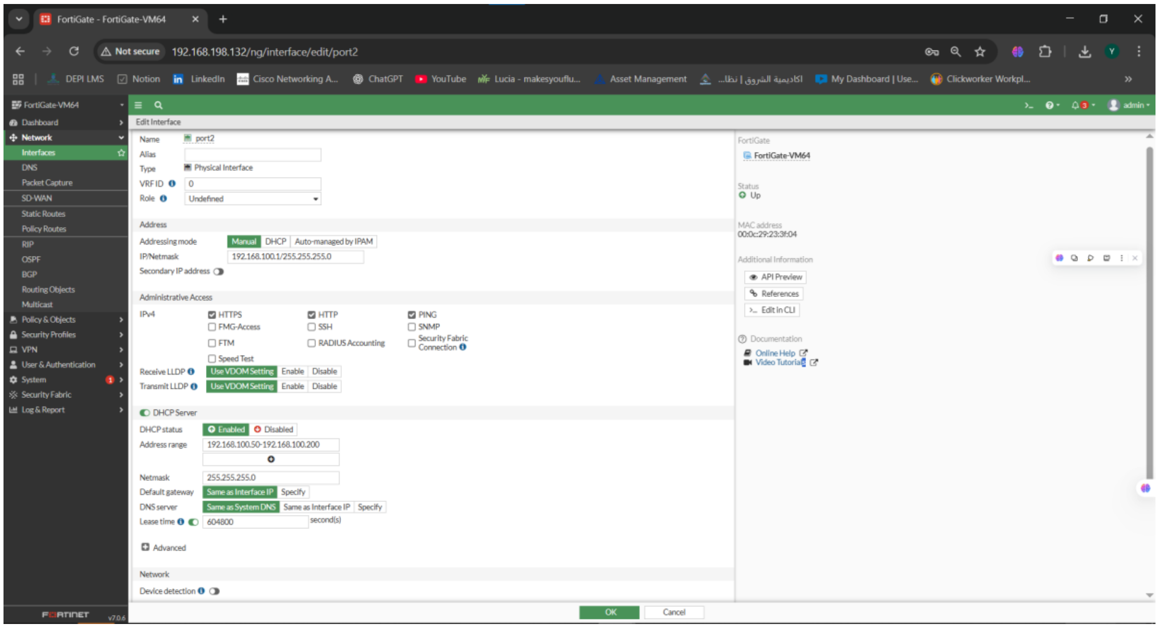


Figure 4 – (Port2 (LAN) Configuration) Port2 configured as the internal LAN interface with IP 192.168.100.1/24

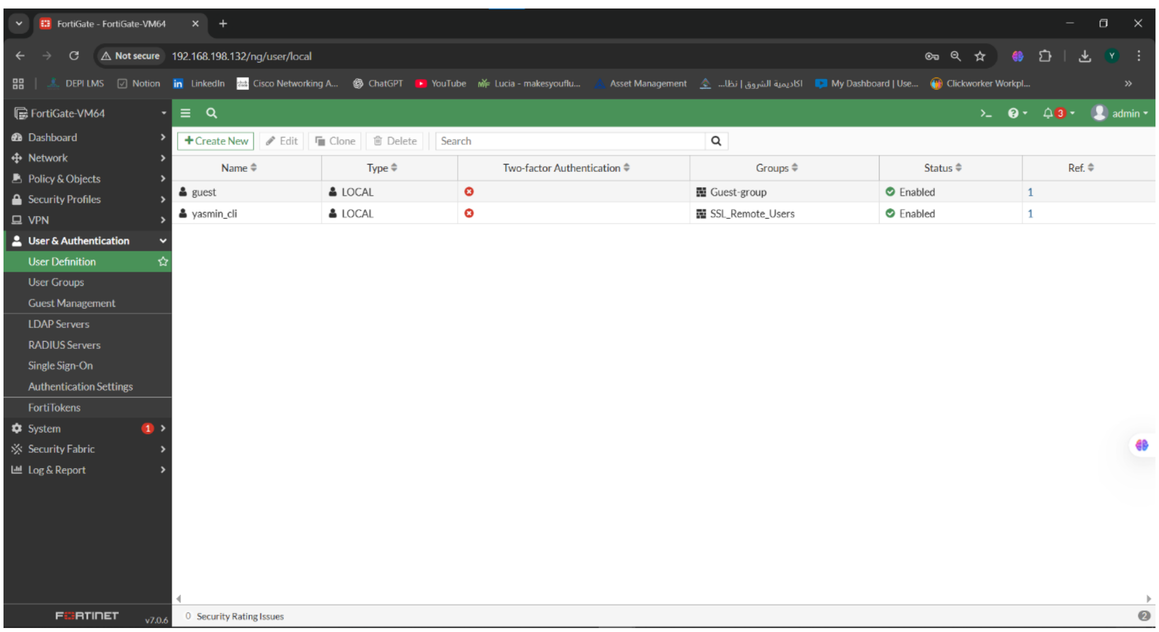


Figure 5 – (Local User (yasmin_cli)) Shows creation of the local user used for SSL-VPN authentication

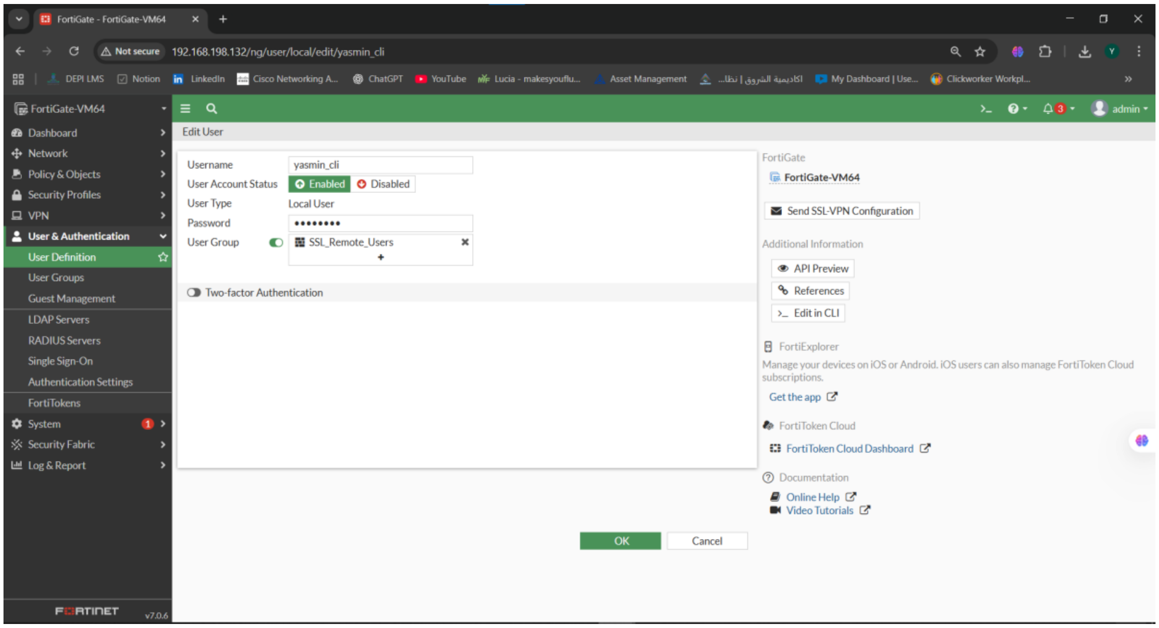


Figure 6 –(Local User Details) Displays full configuration of the user yasmin_cli, including password and status

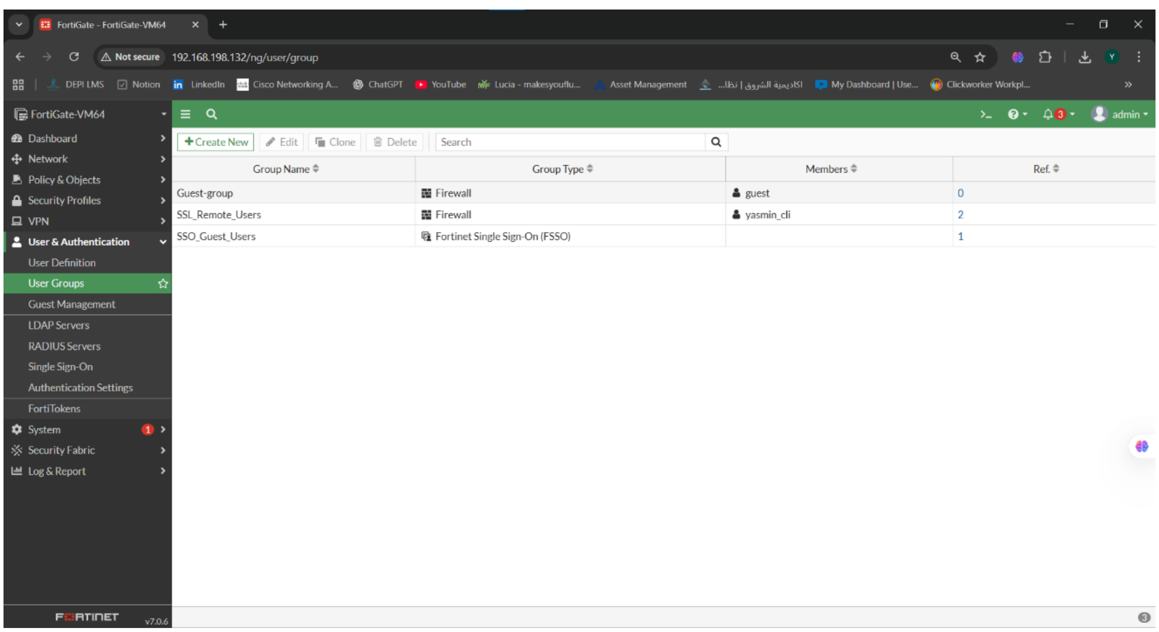


Figure 7 – (SSL_Remote_Users) Group User group created to manage SSL-VPN remote access user

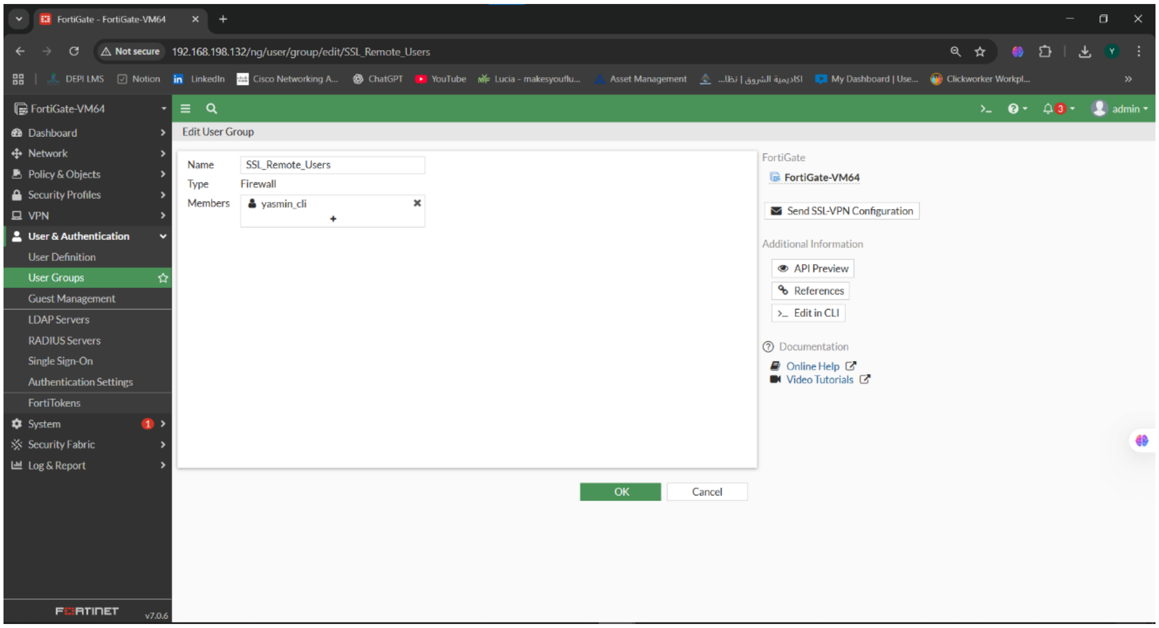


Figure 8 – (Group Members) Shows the user yasmin_cli added to the SSL_Remote_Users group

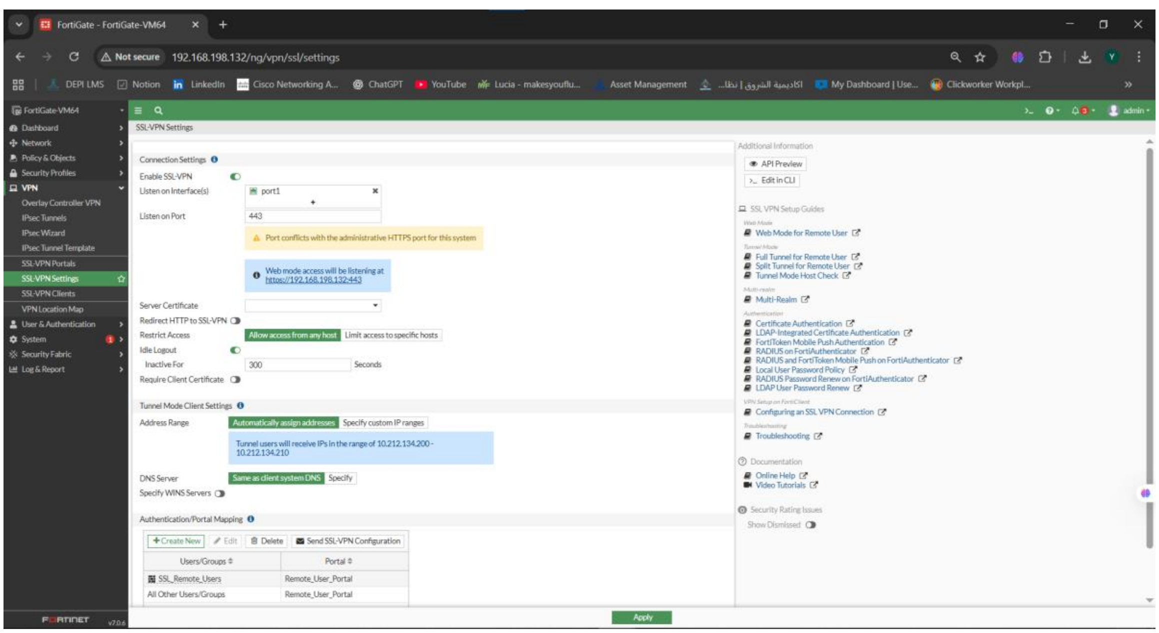


Figure 9 – (SSL-VPN Settings (Main Page)) SSL-VPN enabled on port1 with tunnel mode and address pool configuration.

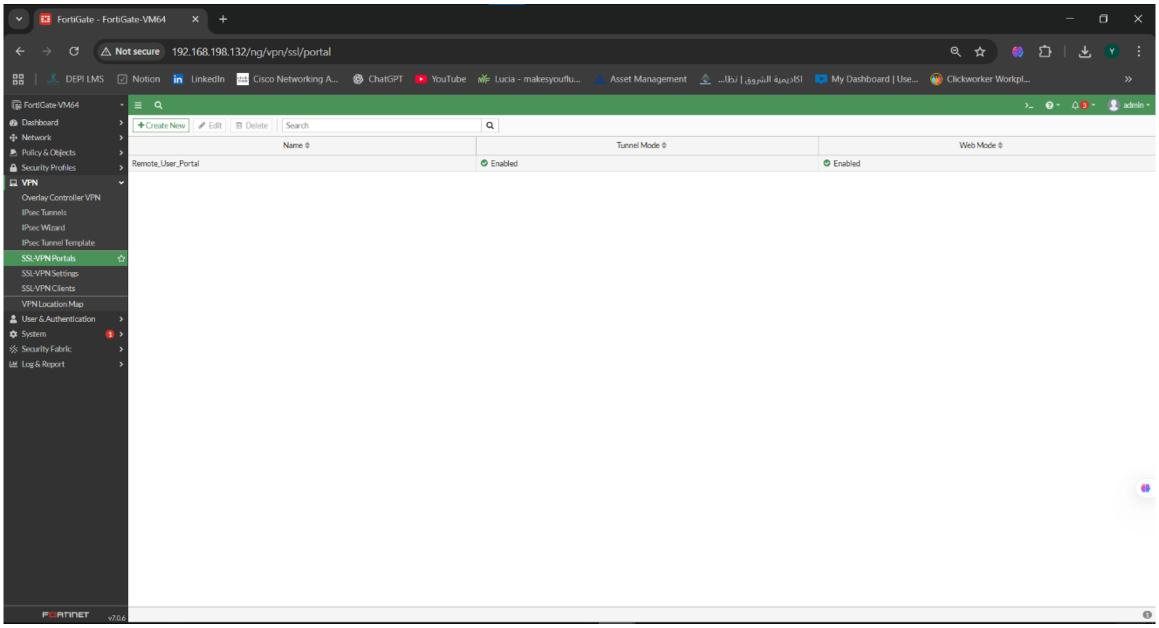


Figure 10 – (SSL-VPN Portal (Remote_User_Portal)) Portal settings for remote users including split tunneling and routing options.

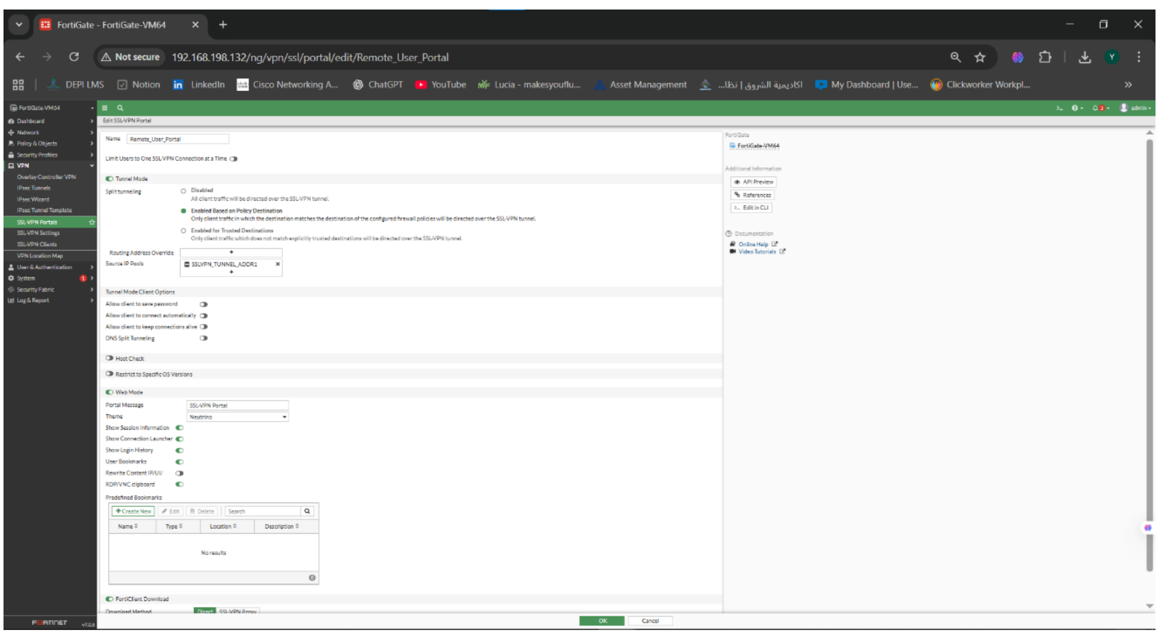


Figure 11 - SSL-VPN Portal Configuration (Remote_User_Portal) with Split-Tunneling Enabled and Routing Address Override Set to SSLVPN_TUNNEL_ADDR1

Figure 11 → This configuration defines the SSL-VPN user portal settings, including tunnel mode behavior, split tunneling based on policy destination, and the IP pool assigned to remote users

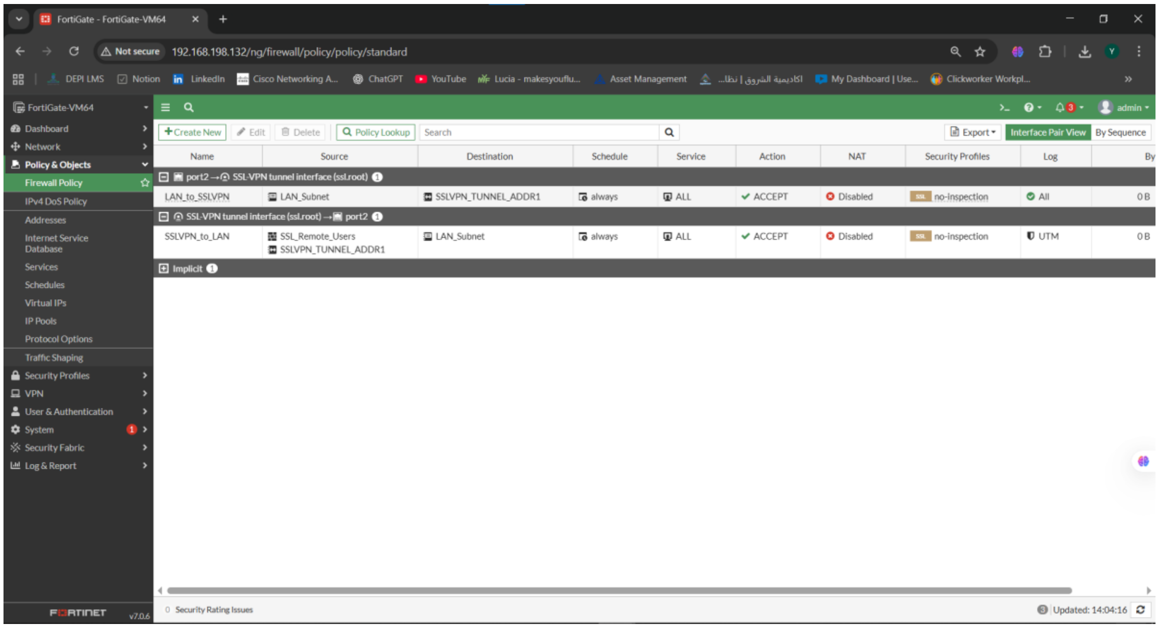


Figure 12 – (Firewall Policies Overview) Overview of the SSL-VPN firewall policies, including LAN_to_SSLVPN and SSLVPN_to_LAN rules.

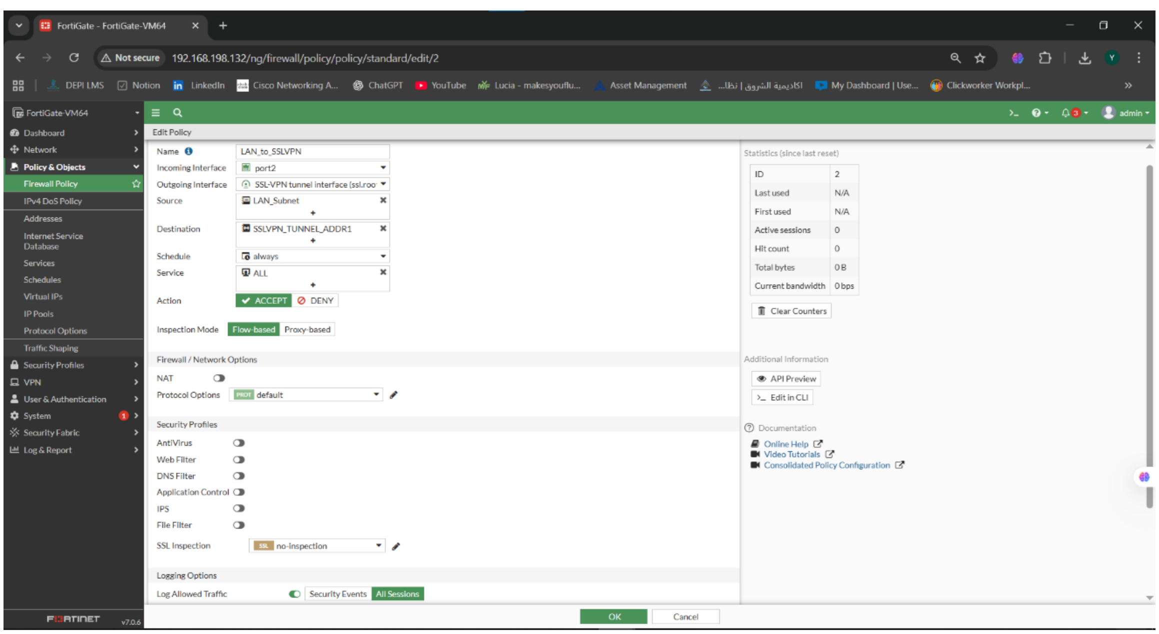


Figure 13 – (Firewall Policy: LAN → SSLVPN) firewall Policy allowing LAN_Subnet to communicate with SSLVPN_TUNNEL_ADDR1 through port2 → ssl.root

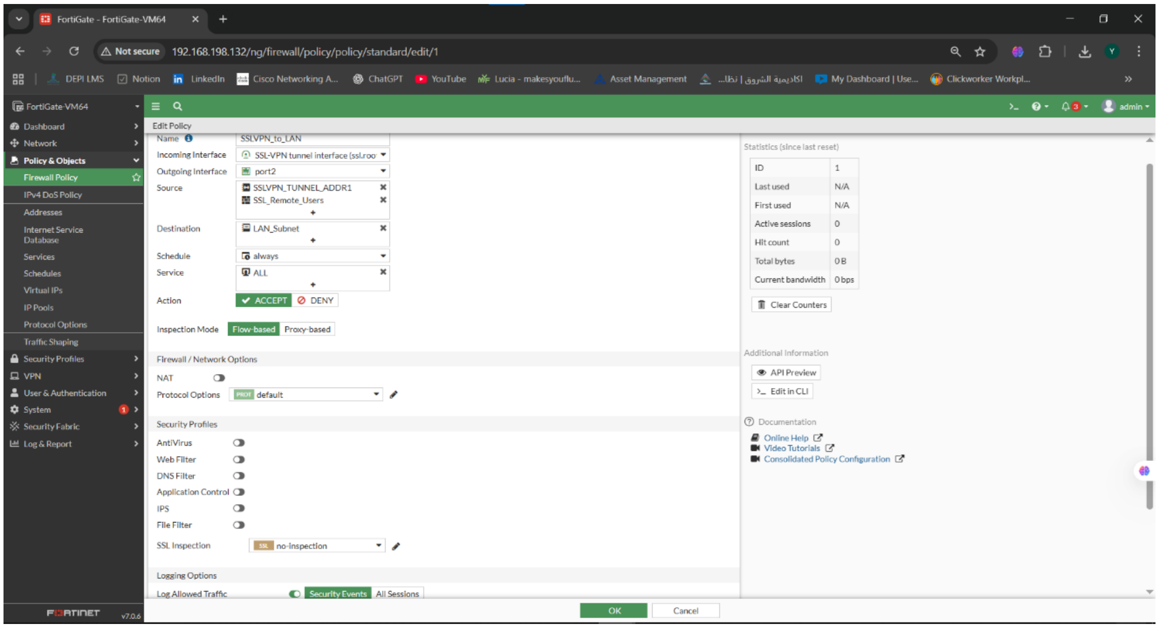


Figure 14 – (Firewall Policy) SSLVPN → LAN Policy allowing SSL-VPN users to access the LAN subnet

4. Test Results

Due to the limitations of the FortiGate VM (no SSL-VPN license),

VPN connection testing could not be performed in Week 1.

However, all required SSL-VPN configuration steps were completed successfully, including:

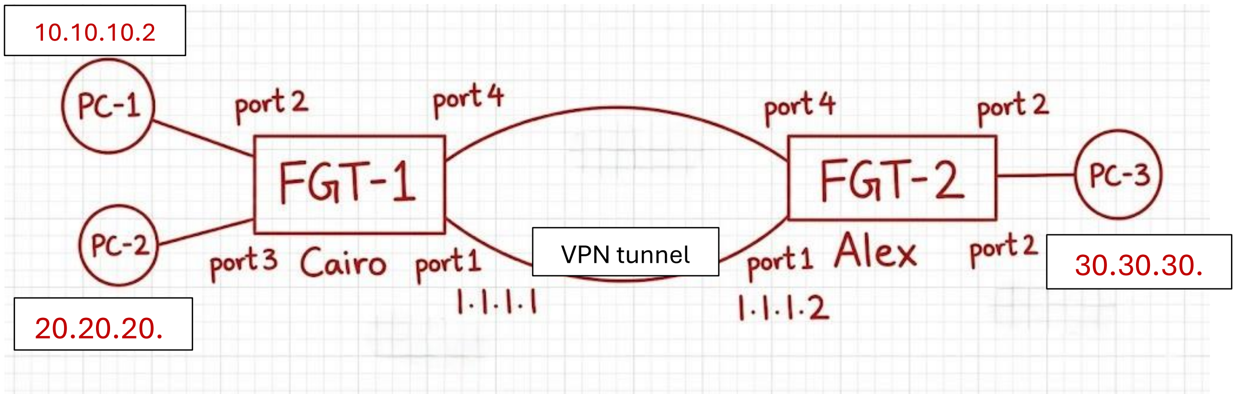
- Interface configuration
- Local user creation
- User group assignment
- SSL-VPN portal setup
- Tunnel IP range configuration
- Firewall policies for SSL-VPN traffic

5. Conclusion

All SSL-VPN configurations required for Week 1 have been successfully completed, including interfaces, users, groups, SSL-VPN settings, portals, and firewall policies. Connection testing is not required in this phase.

IPsec VPN Configuration

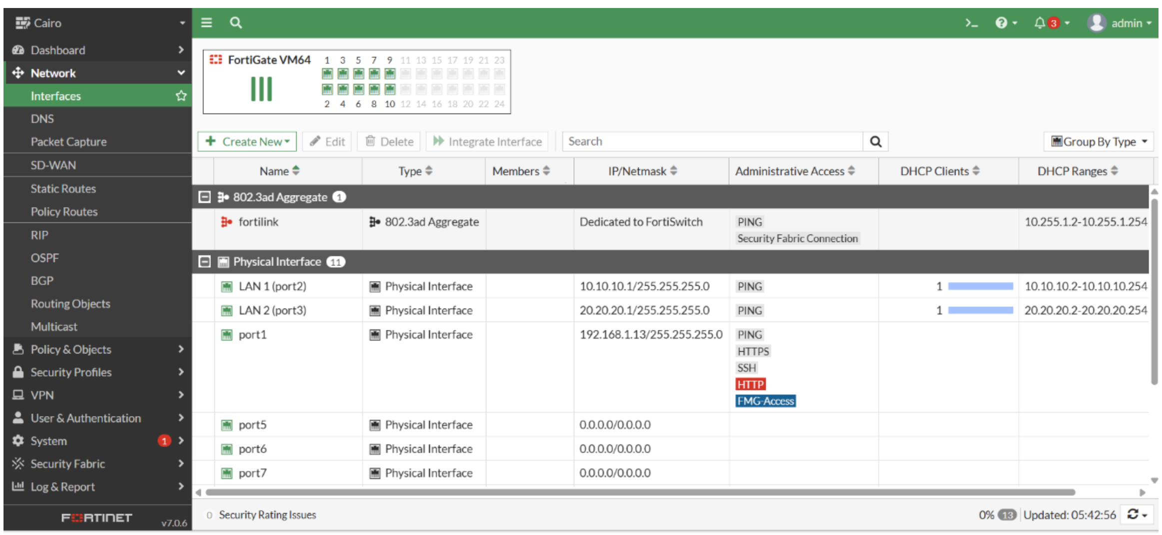
We established this topology:



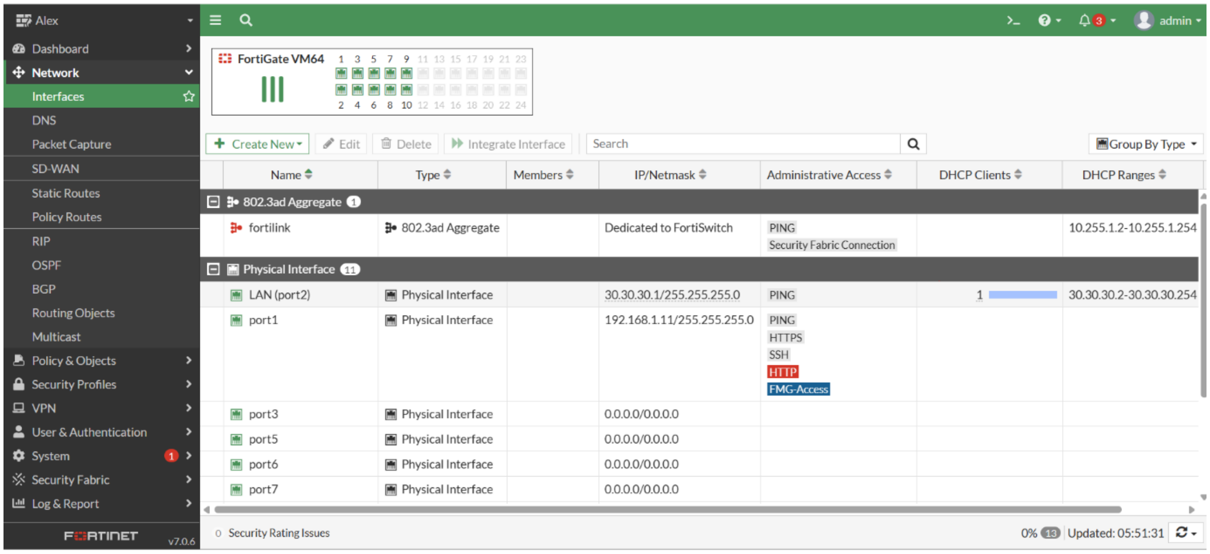
1.Configuration:

1. Configure the interfaces and make port 1 is the management port

Cairo:

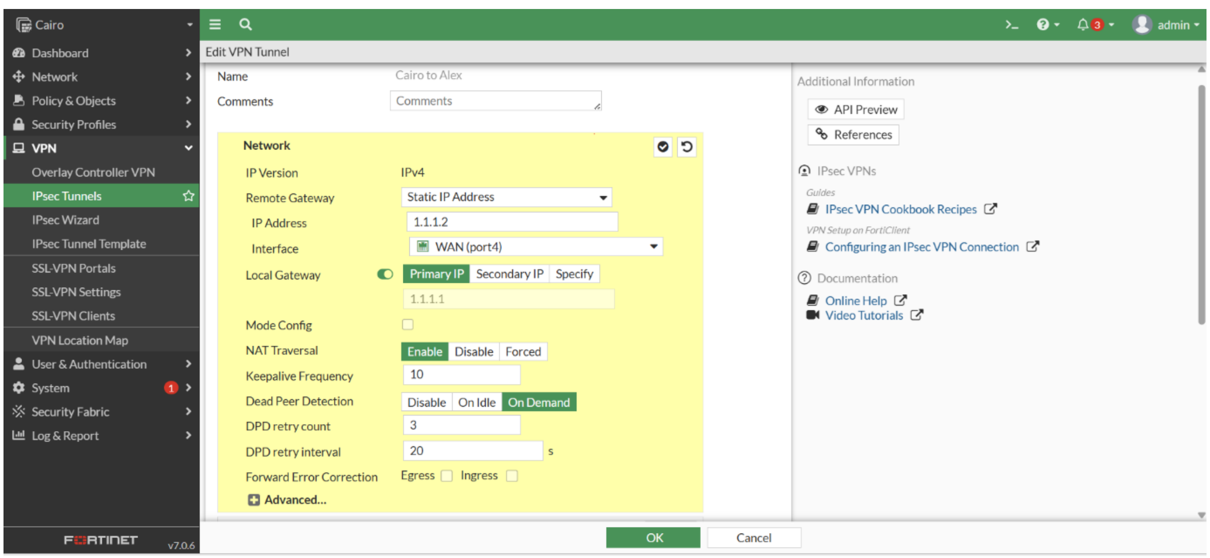


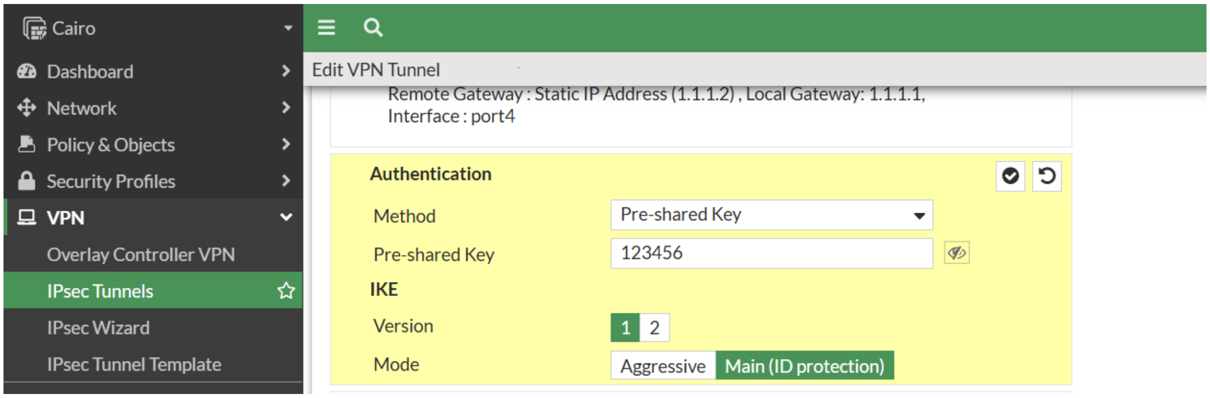
Alex:

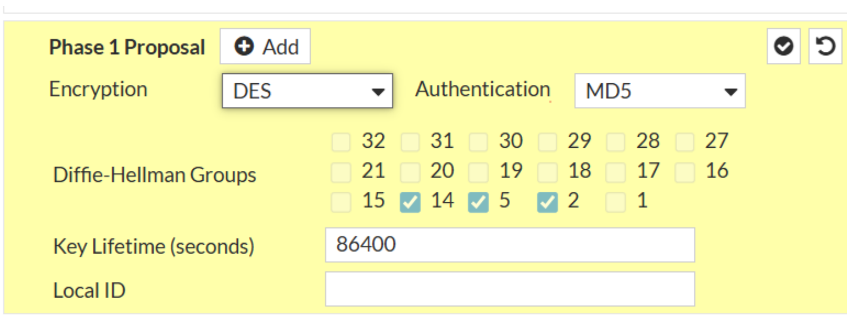


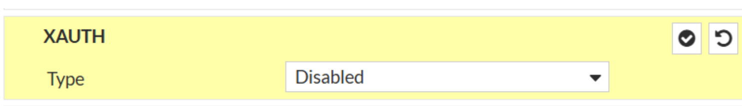
2.Configure the vpn tunnel between the two firewalls and choose traffic selector from the network 10.10.10.0 to 30.30.30.0 in Cairo firewall and reverse in the Alex firewall

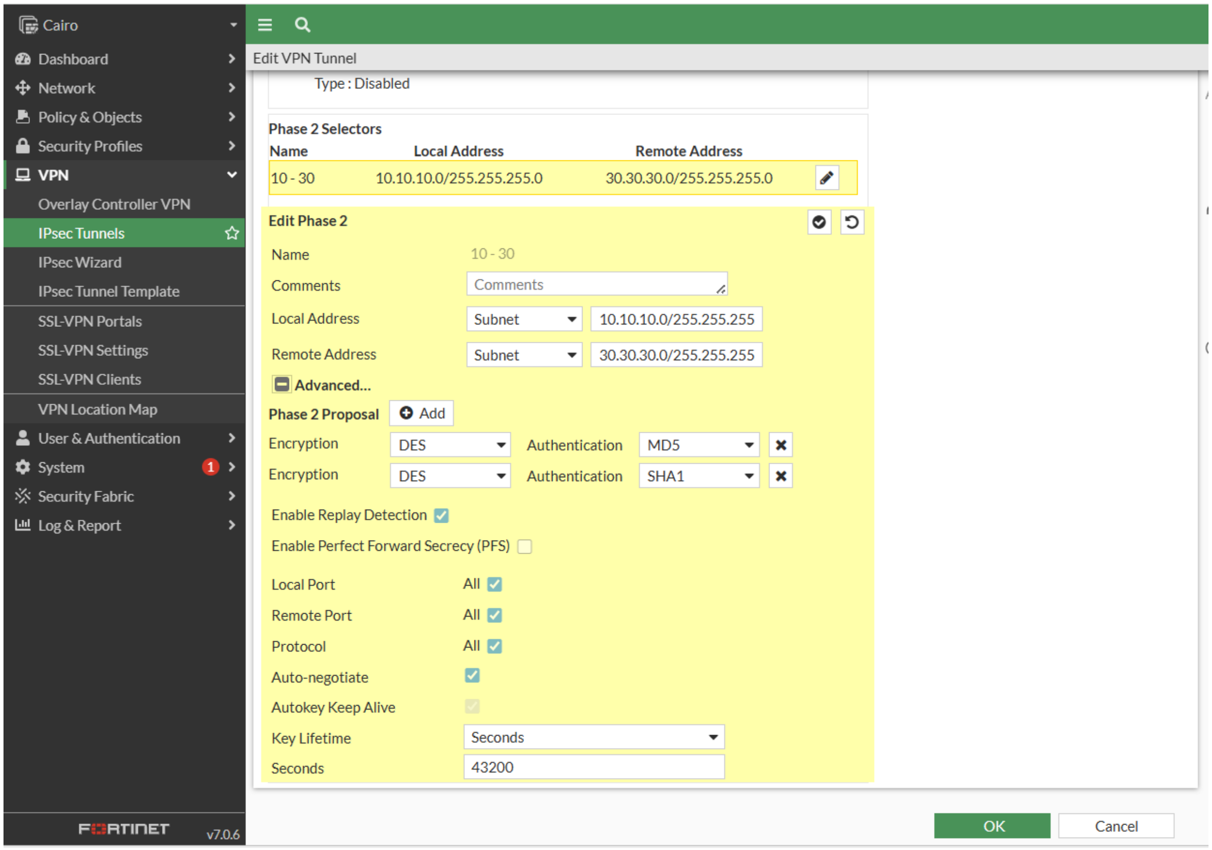
Cairo:




Cairo
Dashboard
Network
Policy & Objects
Security Profiles
VPN
Overlay Controller VPN
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
Edit VPN Tunnel
Remote Gateway : Static IP Address (1.1.1.2) , Local Gateway: 1.1.1.1,
Interface : port4
Authentication
Method
Pre-shared Key
Pre-shared Key
123456
IKE
Version
1
2
Mode
Aggressive
Main (ID protection)


Phase 1 Proposal
Add
Encryption
DES
Authentication
MD5
Diffie-Hellman Groups
32 31 30 29 28 27
21 20 19 18 17 16
15 14 5 2 1
Key Lifetime (seconds)
86400
Local ID


XAUTH
Type
Disabled



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VPN
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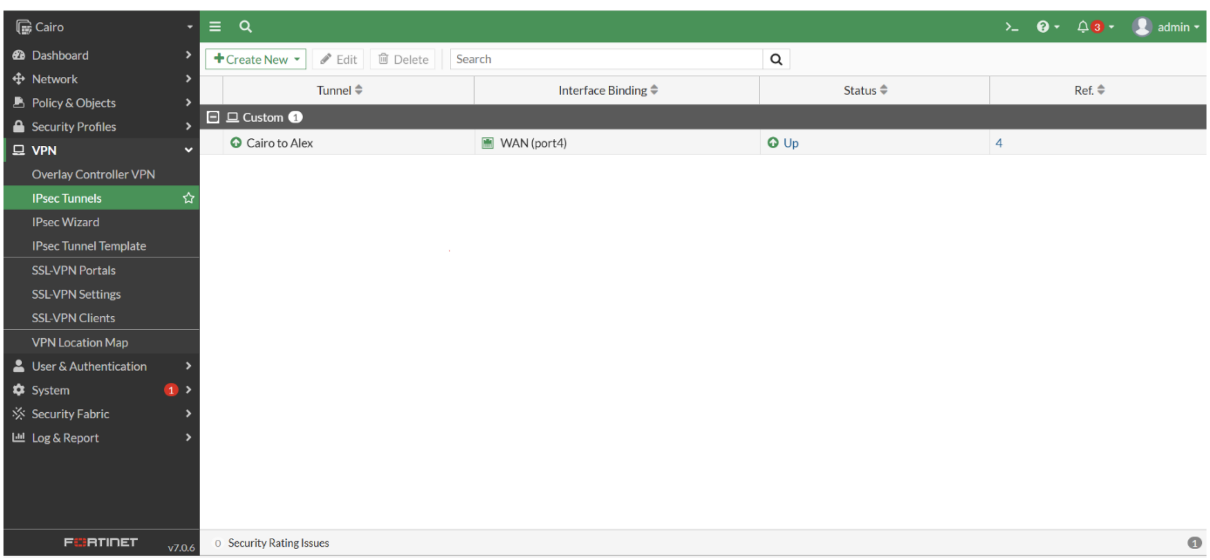
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
v7.0.6
Edit VPN Tunnel
Type : Disabled
Phase 2 Selectors
Name
Local Address
Remote Address
10 - 30
10.10.10.0/255.255.255.0
30.30.30.0/255.255.255.0
Edit Phase 2
Name
10 - 30
Comments
Local Address
Subnet
10.10.10.0/255.255.255
Remote Address
Subnet
30.30.30.0/255.255.255
Advanced...
Phase 2 Proposal
Add
Encryption
DES
Authentication
MD5
Encryption
DES
Authentication
SHA1
Enable Replay Detection
Enable Perfect Forward Secrecy (PFS)
Local Port
All
Remote Port
All
Protocol
All
Auto-negotiate
Autokey Keep Alive
Key Lifetime
Seconds
Seconds
43200
OK
Cancel



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IPsec Tunnel Template
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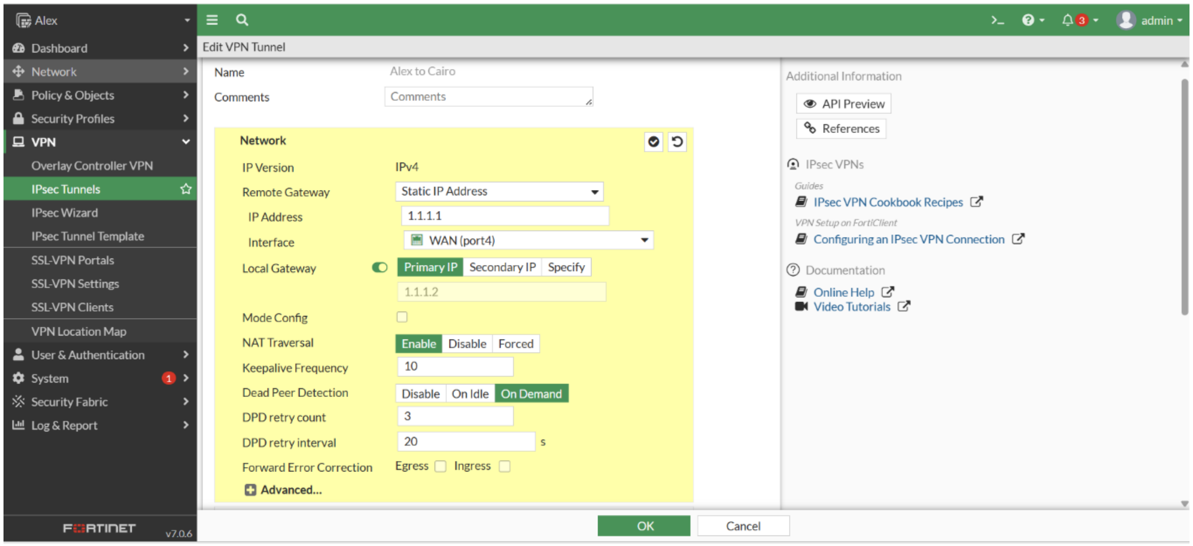
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
v7.0.6
admin
Create New
Edit
Delete
Search
Tunnel
Interface Binding
Status
Ref.
Custom
Cairo to Alex
WAN (port4)
Up
4
0 Security Rating Issues

Alex: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IPsec Tunnels
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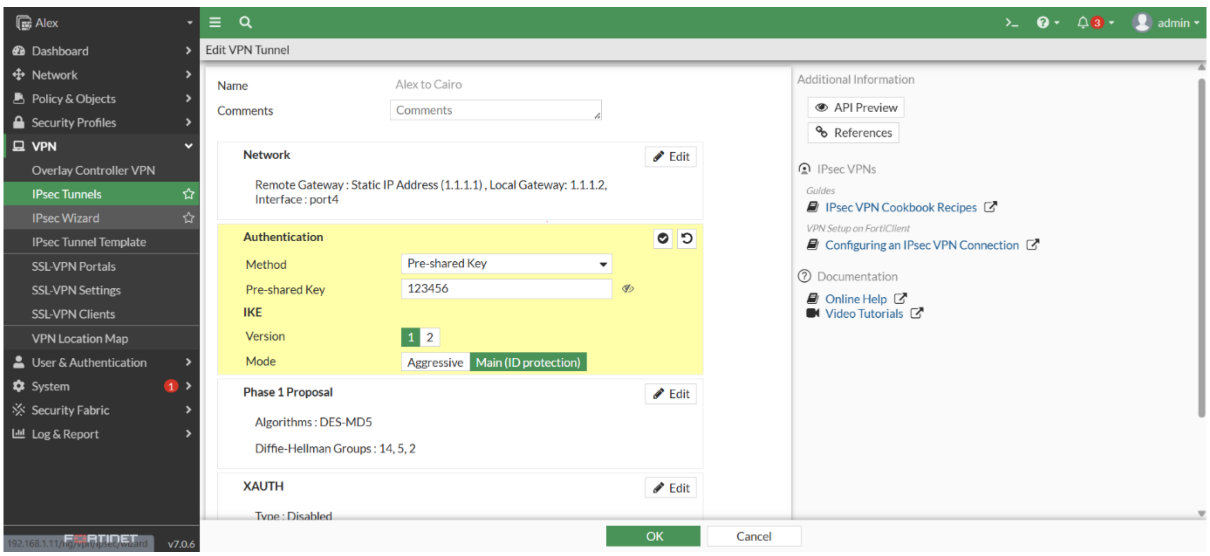
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
FORTINET v7.0.6
admin
Edit VPN Tunnel
Name Alex to Cairo
Comments Comments
Network
IP Version IPv4
Remote Gateway Static IP Address
IP Address 1.1.1.1
Interface WAN (port4)
Local Gateway Primary IP Secondary IP Specify
1.1.1.2
Mode Config
NAT Traversal Enable Disable Forced
Keepalive Frequency 10
Dead Peer Detection Disable On Idle On Demand
DPD retry count 3
DPD retry interval 20 s
Forward Error Correction Egress Ingress
Advanced...
OK
Cancel
Additional Information
API Preview
References
IPsec VPNs
Guides
IPsec VPN Cookbook Recipes
VPN Setup on FortiClient
Configuring an IPsec VPN Connection
Documentation
Online Help
Video Tutorials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SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
Security Fabric
Log & Report
v7.0.6
admin
Edit VPN Tunnel
Name Alex to Cairo
Comments Comments
Network
Edit
Remote Gateway : Static IP Address (1.1.1.1) , Local Gateway: 1.1.1.2,
Interface : port4
Authentication
Method Pre-shared Key
Pre-shared Key 123456
IKE
Version 1 2
Mode Aggressive Main (ID protection)
Phase 1 Proposal
Edit
Algorithms : DES-MD5
Diffie-Hellman Groups : 14, 5, 2
XAUTH
Edit
Type : Disabled
OK
Cancel
Additional Information
API Preview
References
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IPsec VPN Cookbook Recipes
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Configuring an IPsec VPN Connection
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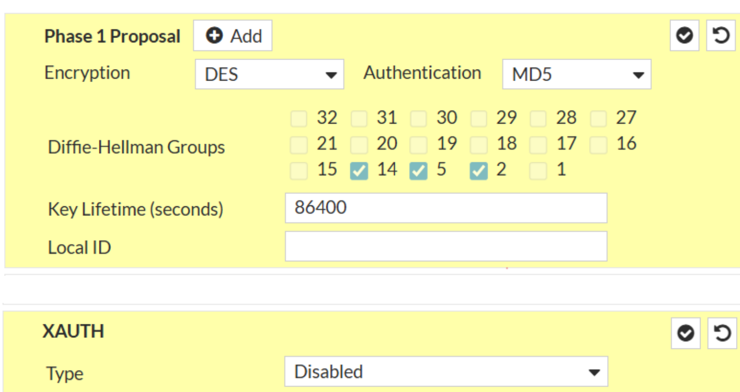
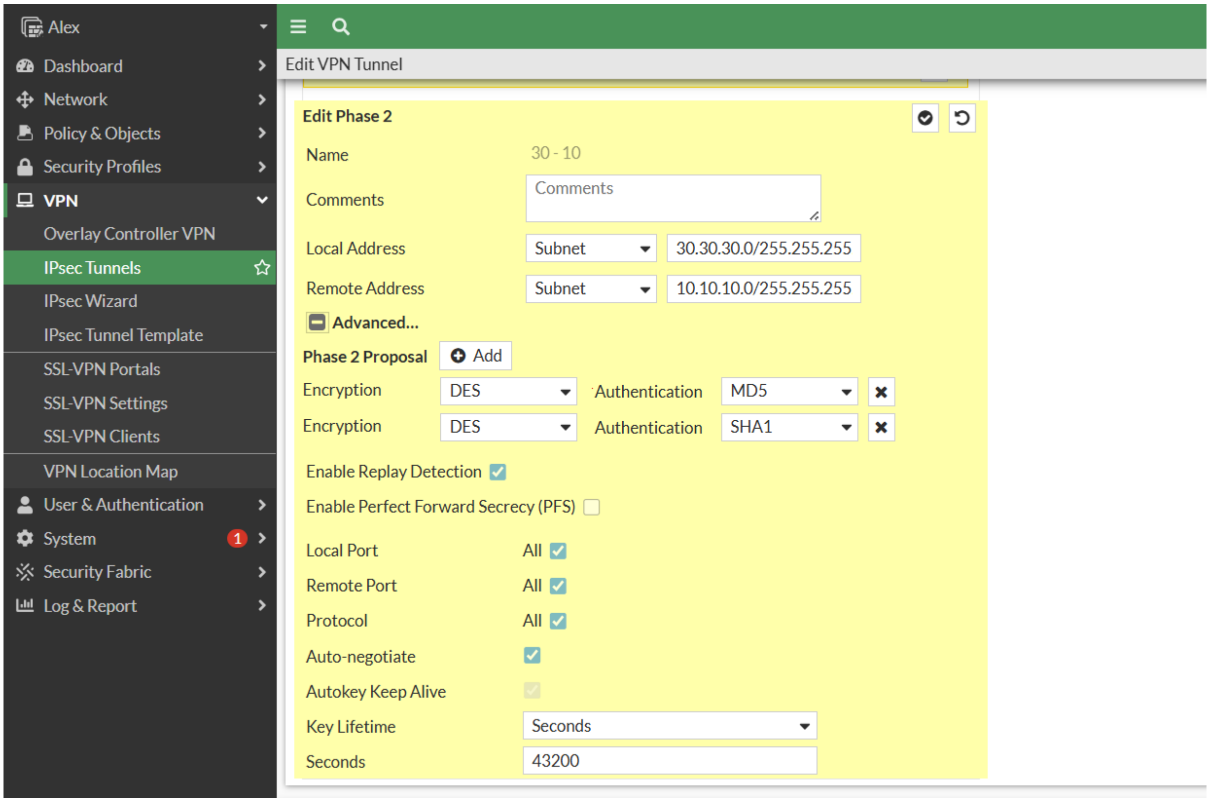
Video Tutorials


Phase 1 Proposal
Add
Encryption DES
Authentication MD5
Diffie-Hellman Groups
32 31 30 29 28 27
21 20 19 18 17 16
15 14 5 2 1
Key Lifetime (seconds) 86400
Local ID
XAUTH
Type Disabled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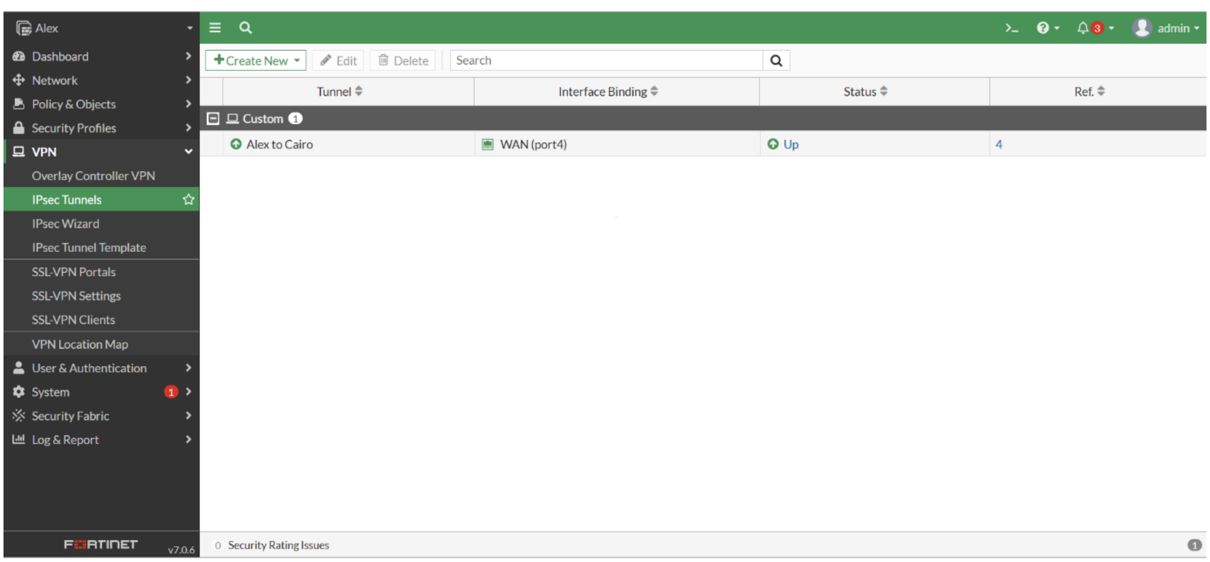
IPsec Tunnels
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
1
Security Fabric
Log & Report
Edit VPN Tunnel
Edit Phase 2
Name
30 - 10
Comments
Comments
Local Address
Subnet
30.30.30.0/255.255.255
Remote Address
Subnet
10.10.10.0/255.255.255
Advanced...
Phase 2 Proposal
Add
Encryption
DES
Authentication
MD5
Encryption
DES
Authentication
SHA1
Enable Replay Detection
Enable Perfect Forward Secrecy (PFS)
Local Port
All
Remote Port
All
Protocol
All
Auto-negotiate
Autokey Keep Alive
Key Lifetime
Seconds
Seconds
43200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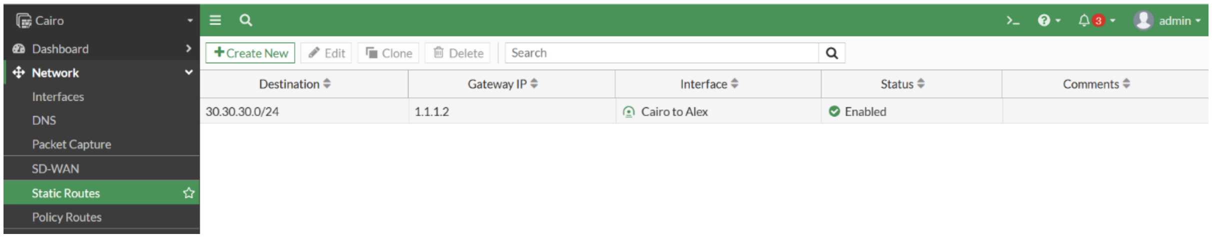
IPsec Wizard
IPsec Tunnel Template
SSL-VPN Portals
SSL-VPN Settings
SSL-VPN Clients
VPN Location Map
User & Authentication
System
1
Security Fabric
Log & Report
FORTINET
v7.0.6
admin
Create New
Edit
Delete
Search
Tunnel
Interface Binding
Status
Ref.
Custom
1
Alex to Cairo
WAN (port4)
Up
4
0
Security Rating Issues

3.Configure the static route

Cairo:



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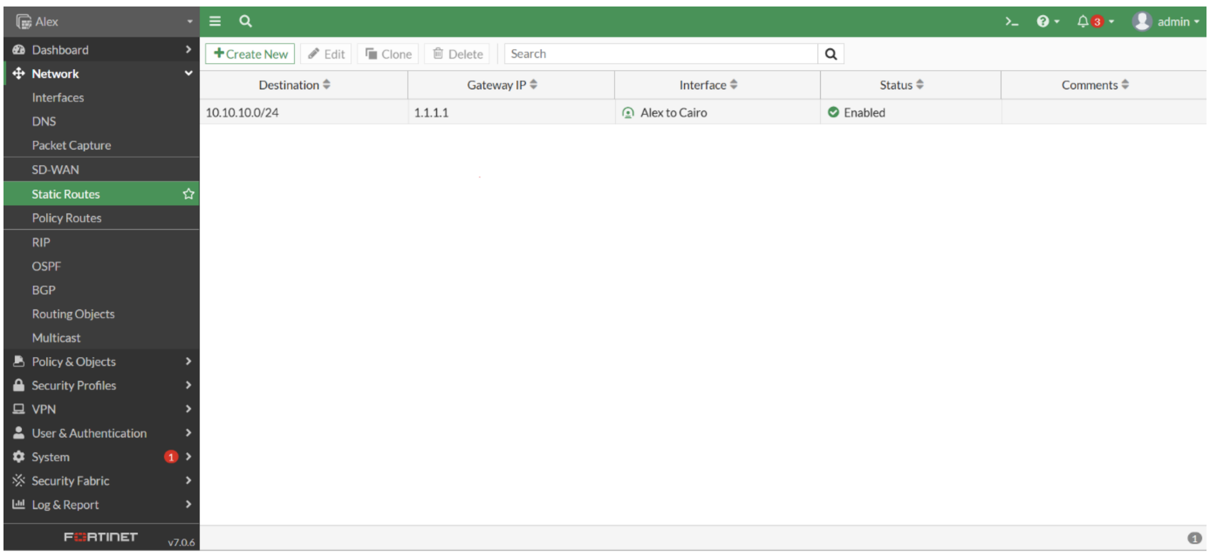
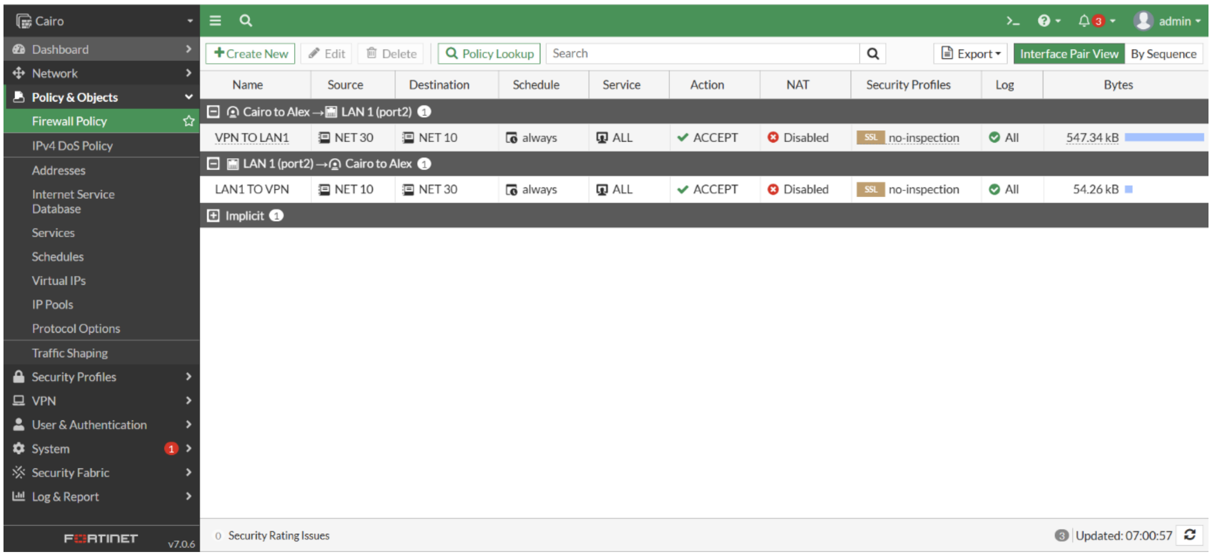


4.Configure the policy between network 10.10.10.0 and 30.30.30.0

Cairo:



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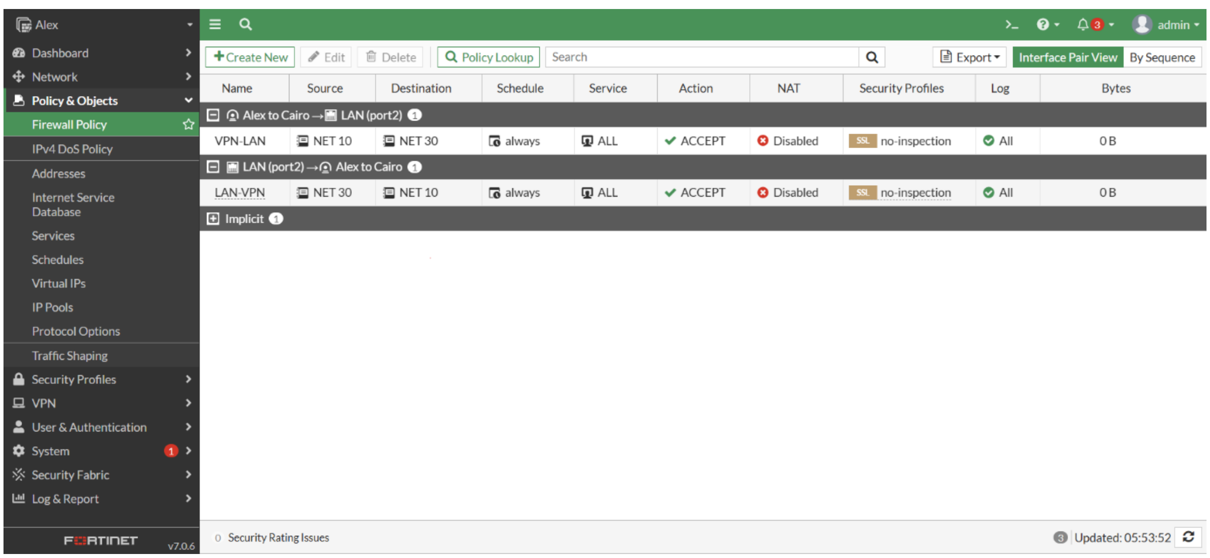
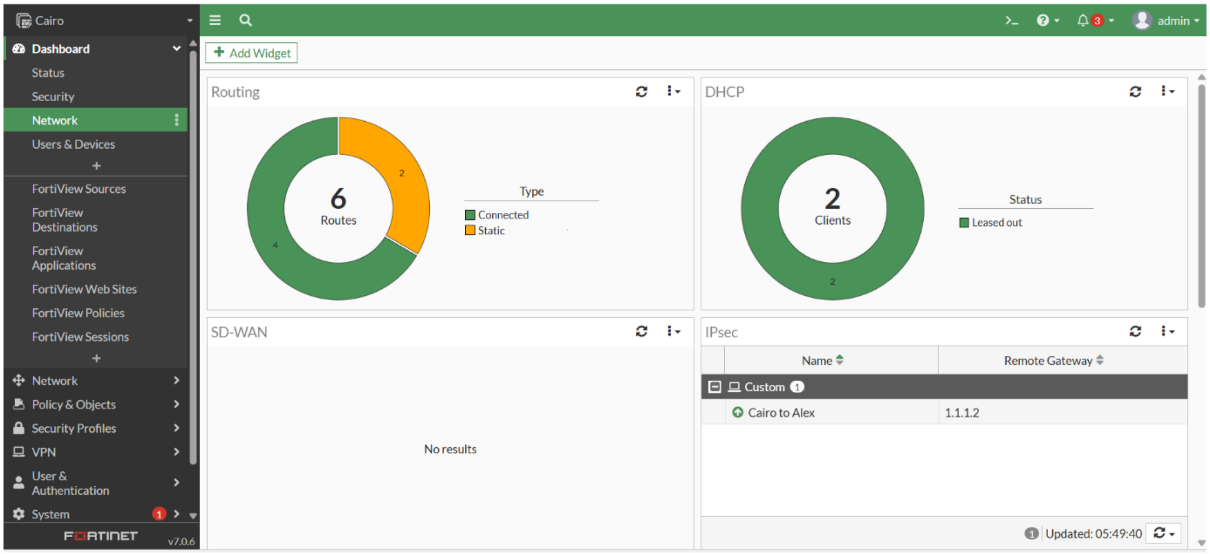
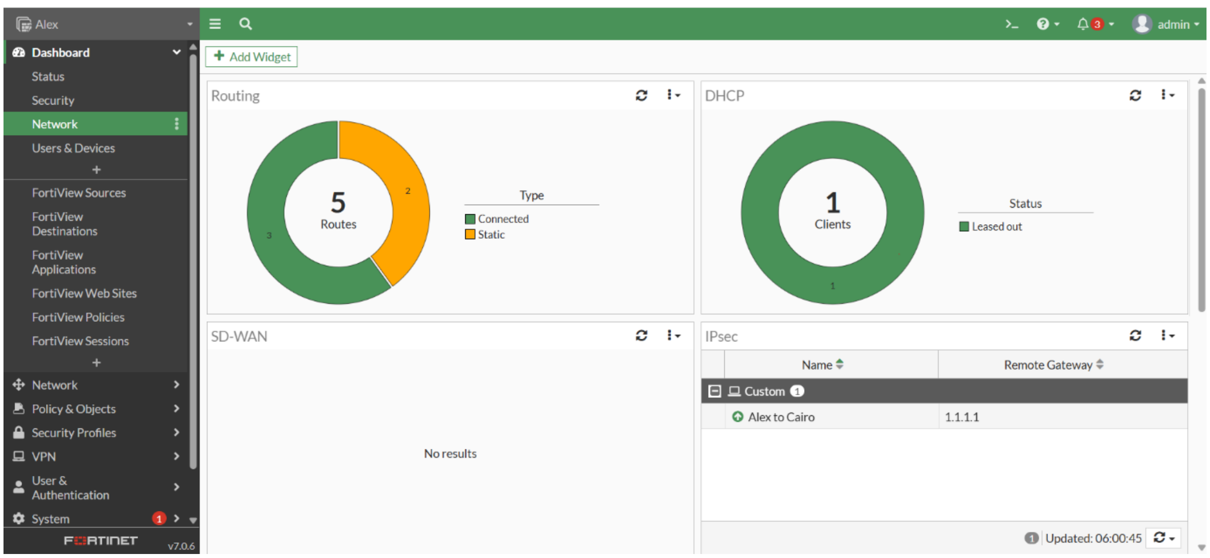


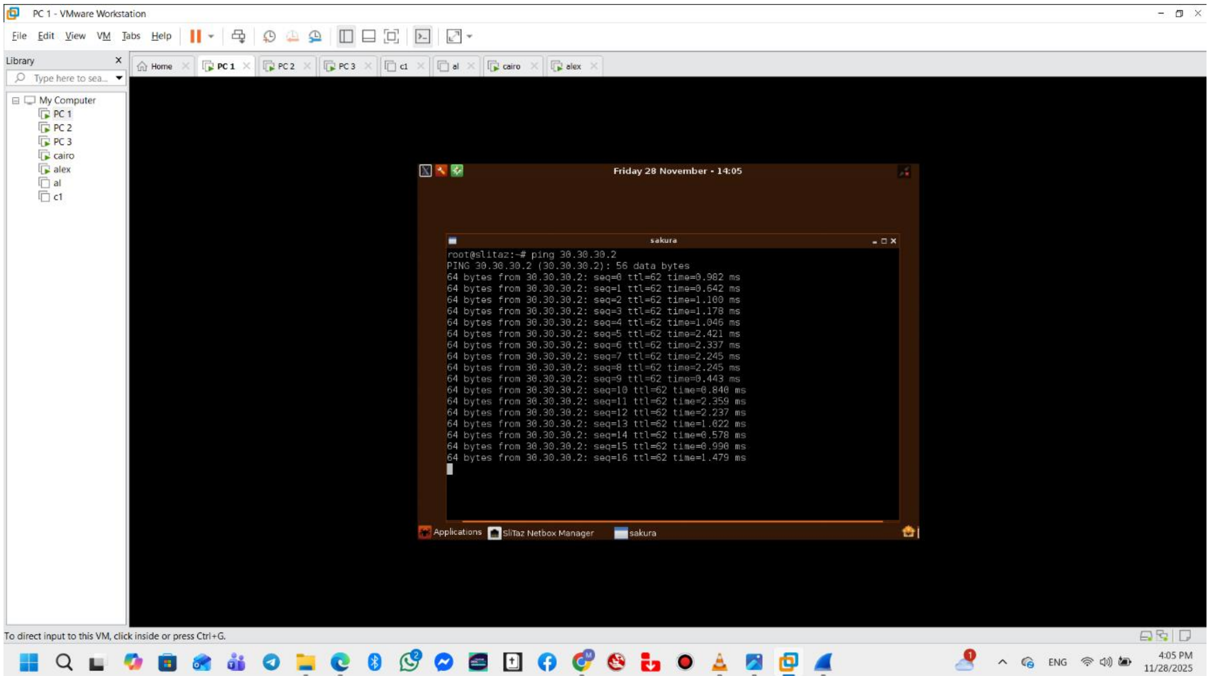
So the Vpn tunnel now is up and running

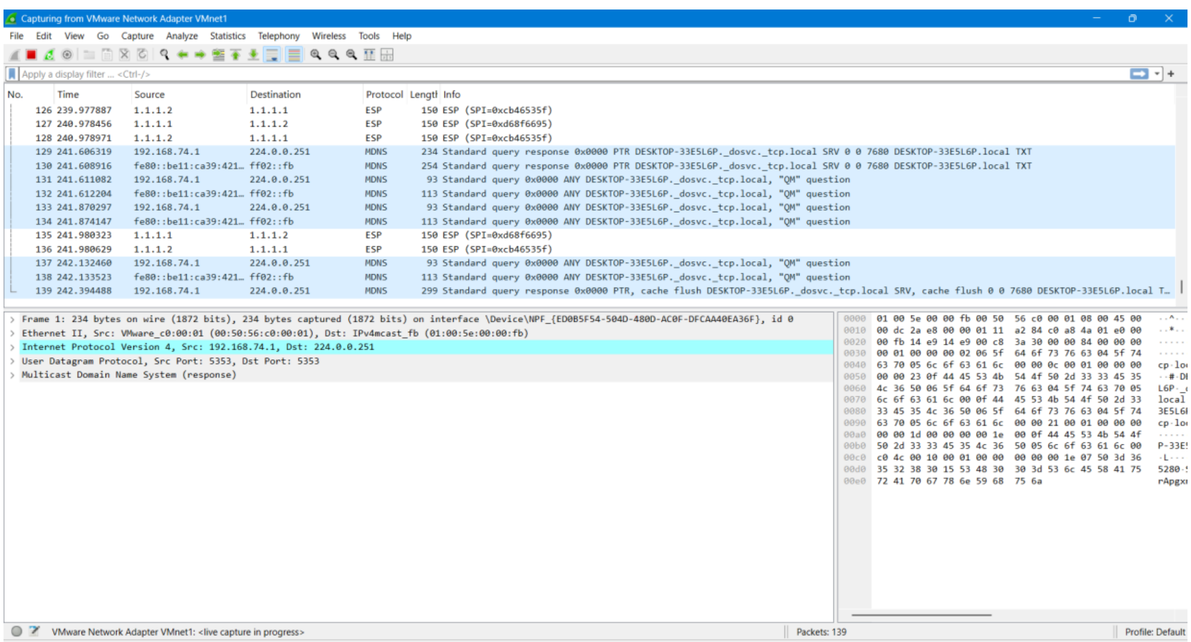

Cairo
Dashboard
Status
Security
Network
Users & Devices
FortiView Sources
FortiView Destinations
FortiView Applications
FortiView Web Sites
FortiView Policies
FortiView Sessions
Network
Policy & Objects
Security Profiles
VPN
User & Authentication
System
v7.0.6
admin
Add Widget
Routing
6
Routes
Type
Connected
Static
DHCP
2
Clients
Status
Leased out
SD-WAN
No results
IPsec
Name
Remote Gateway
Custom
Cairo to Alex
1.1.1.2
Updated: 05:49:40


Alex
Dashboard
Status
Security
Network
Users & Devices
FortiView Sources
FortiView Destinations
FortiView Applications
FortiView Web Sites
FortiView Policies
FortiView Sessions
Network
Policy & Objects
Security Profiles
VPN
User & Authentication
System
v7.0.6
admin
Add Widget
Routing
5
Routes
Type
Connected
Static
DHCP
1
Clients
Status
Leased out
SD-WAN
No results
IPsec
Name
Remote Gateway
Custom
Alex to Cairo
1.1.1.1
Updated: 06:00:45

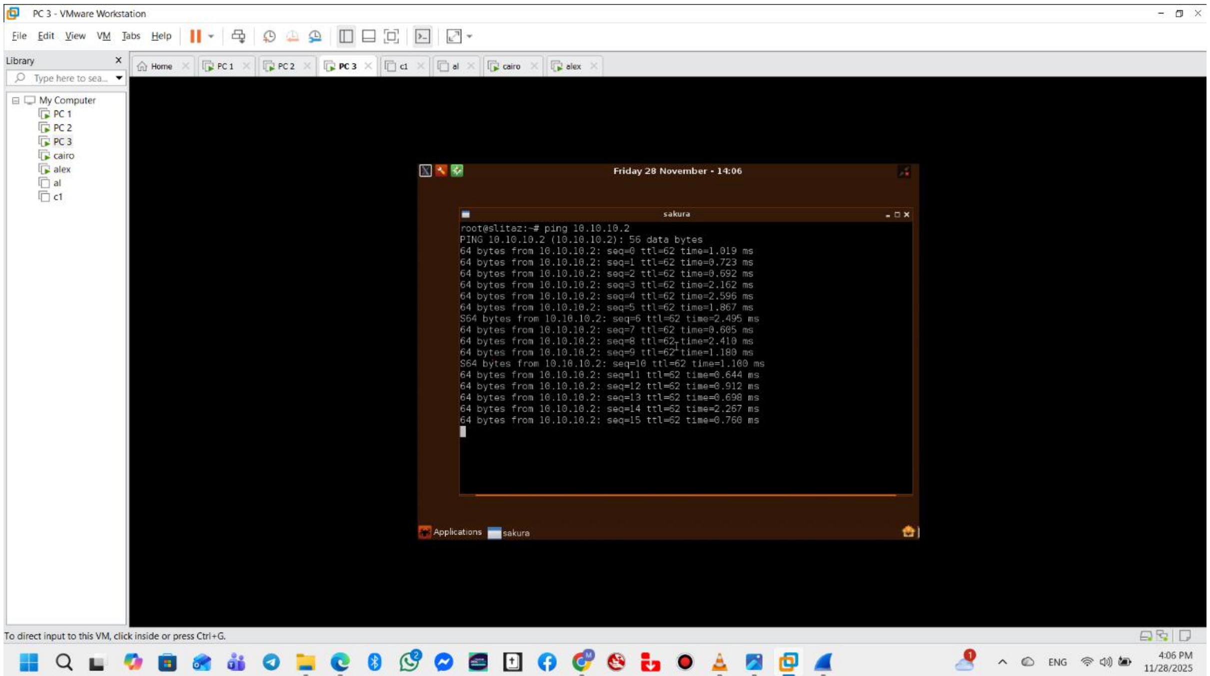
Test Results

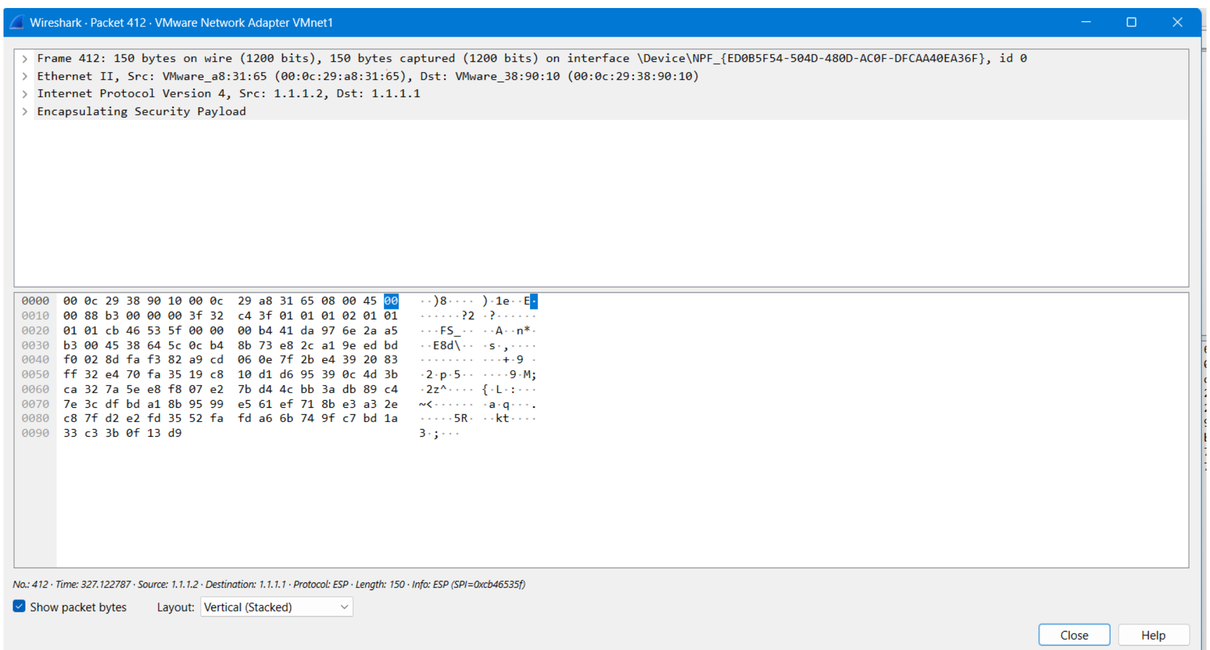
So let us see the result when we ping from PC-1 to PC-3





From PC-3 to PC-1

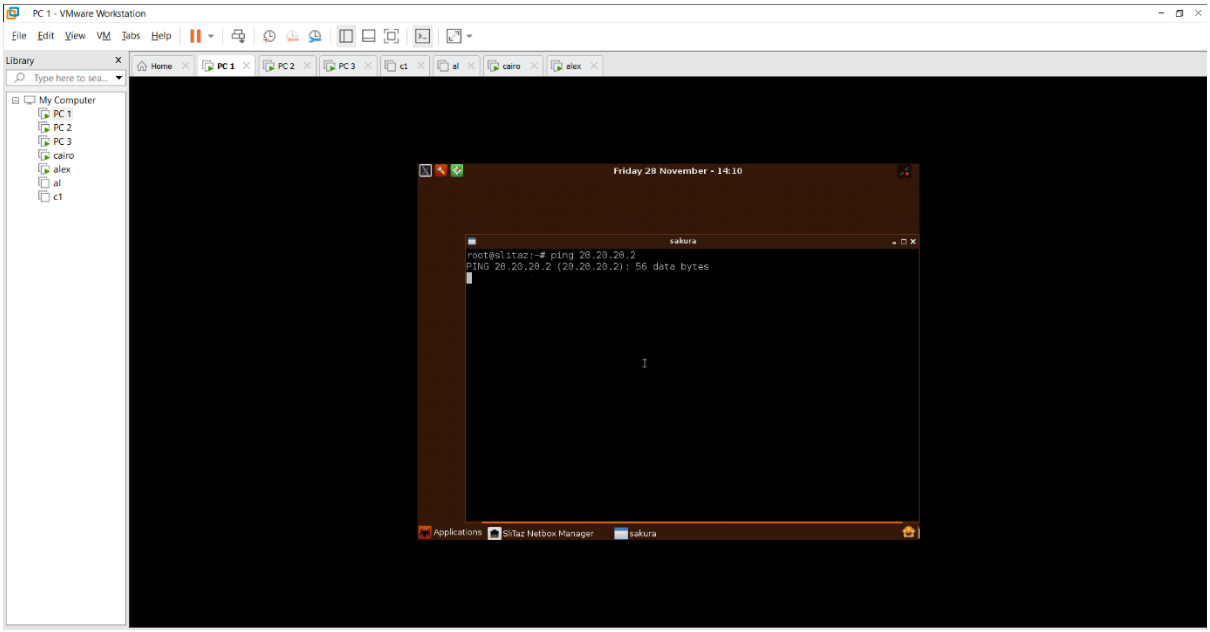




Comment:

The connectivity test results between PC-1 and PC-3 are successful because we established traffic selector from the network 10.10.10.0 to 30.30.30.0

Ping from PC-1 to PC-2



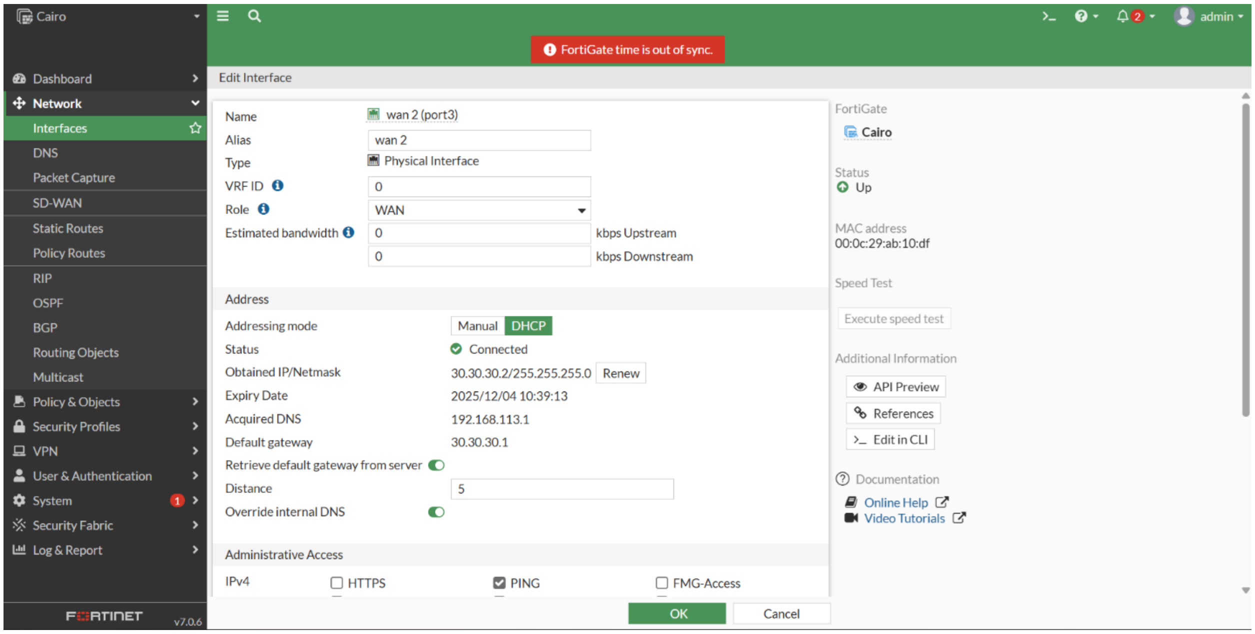
Comment:

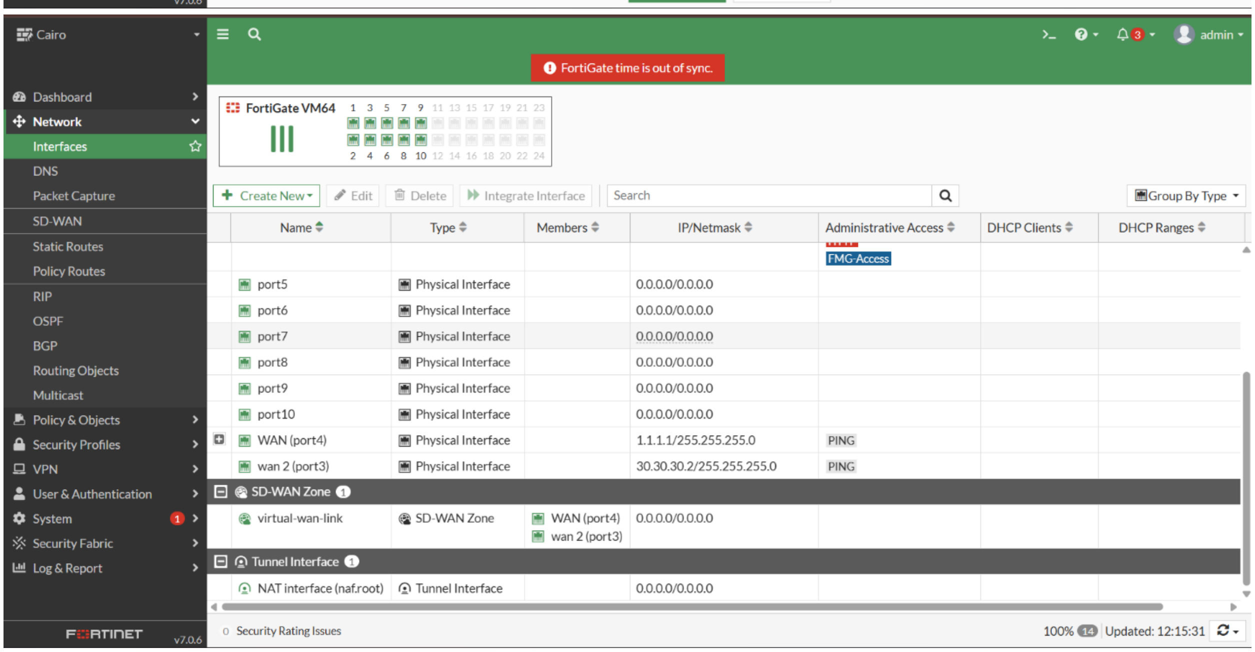
The connectivity test results between PC-1 and PC-2 have failed because there is no traffic selector between network 10 and network 20 so the result is as expected

Fortigate VPN with SD-WAN

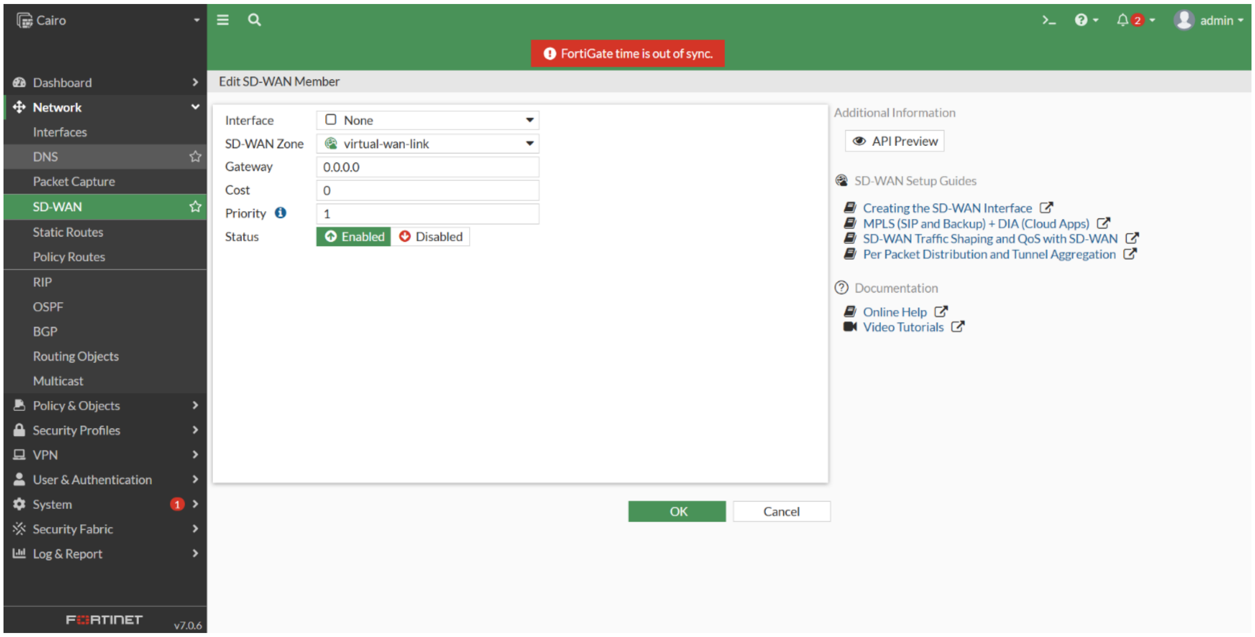
Configuration of SD-WAN on fortigate

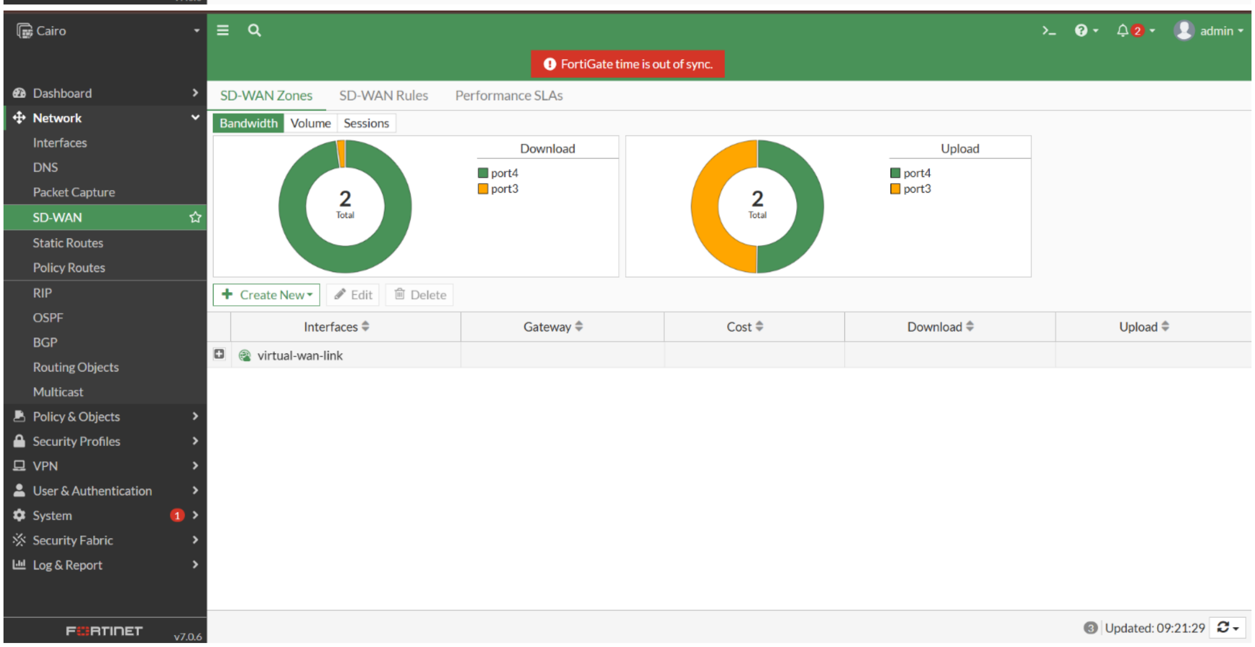
1.Configure the ports to a wan





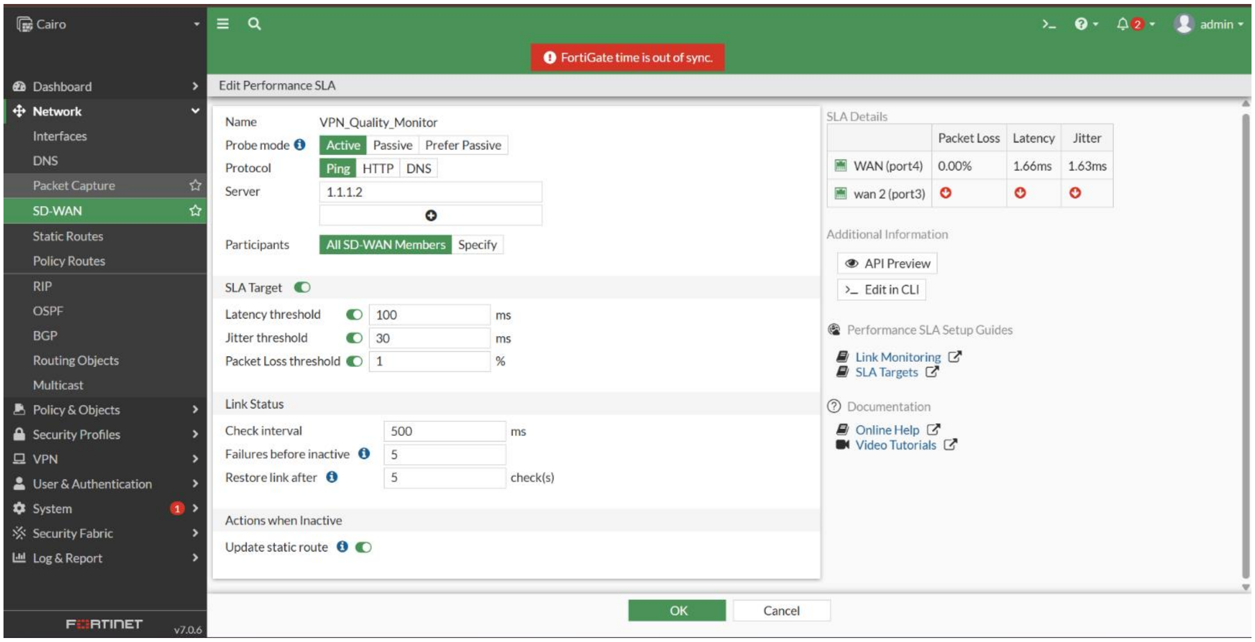
2.Create new SD-WAN member





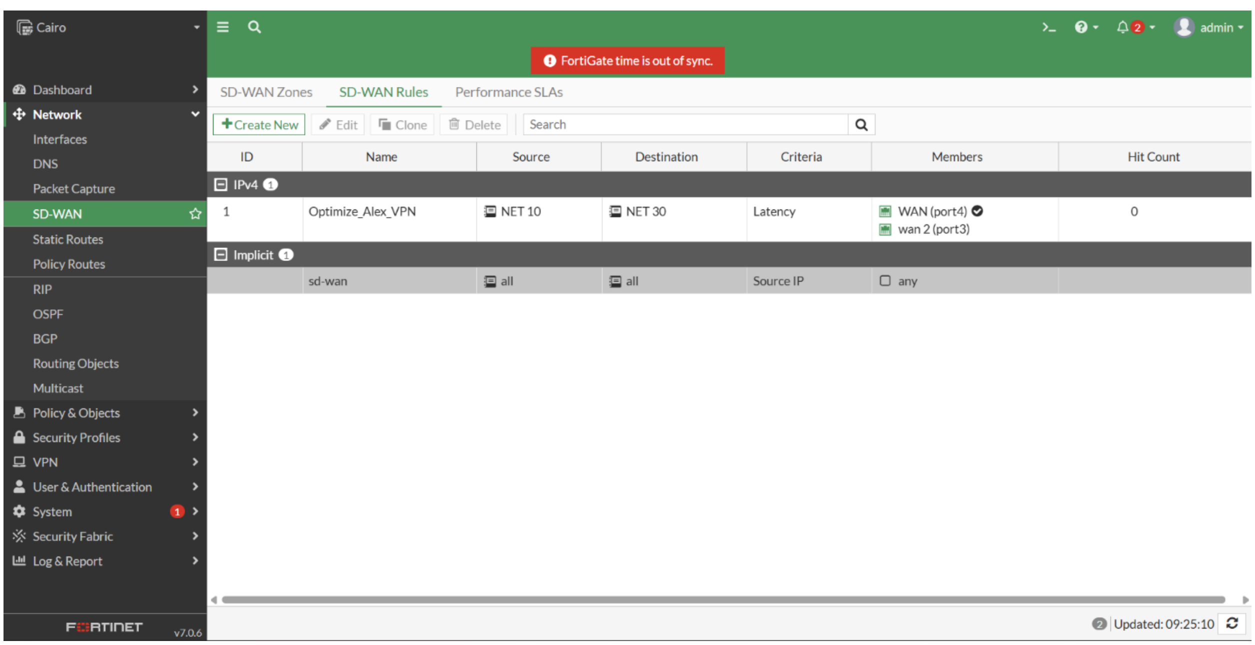
3.Define Link Performance Metrics (Performance SLA)

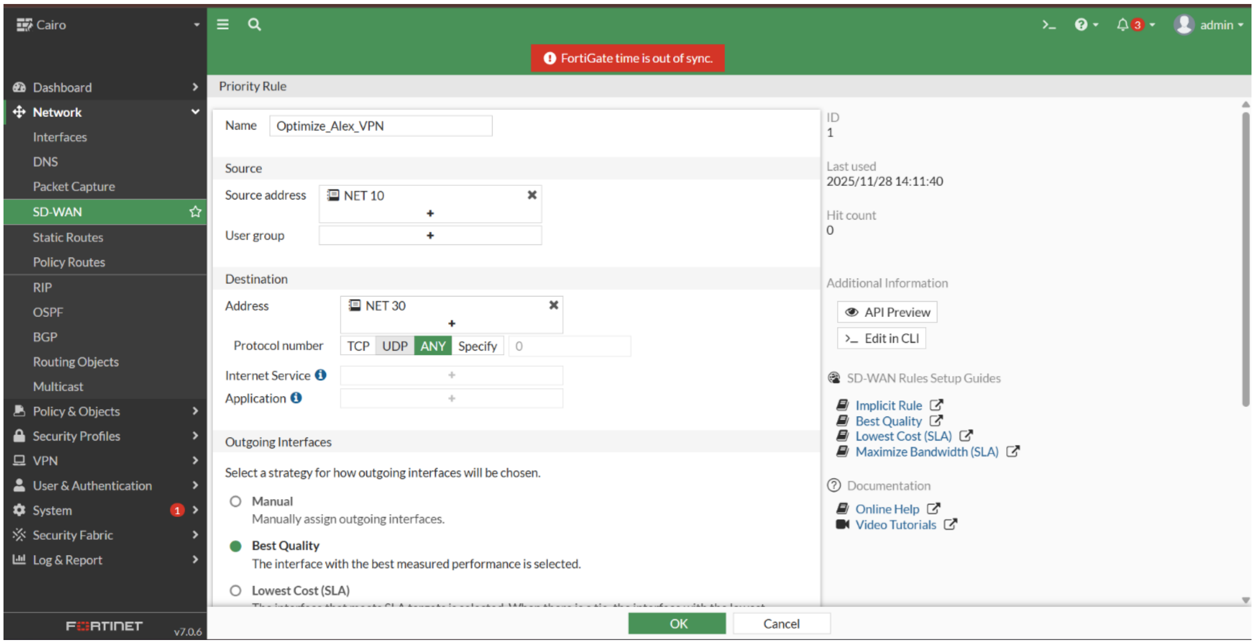
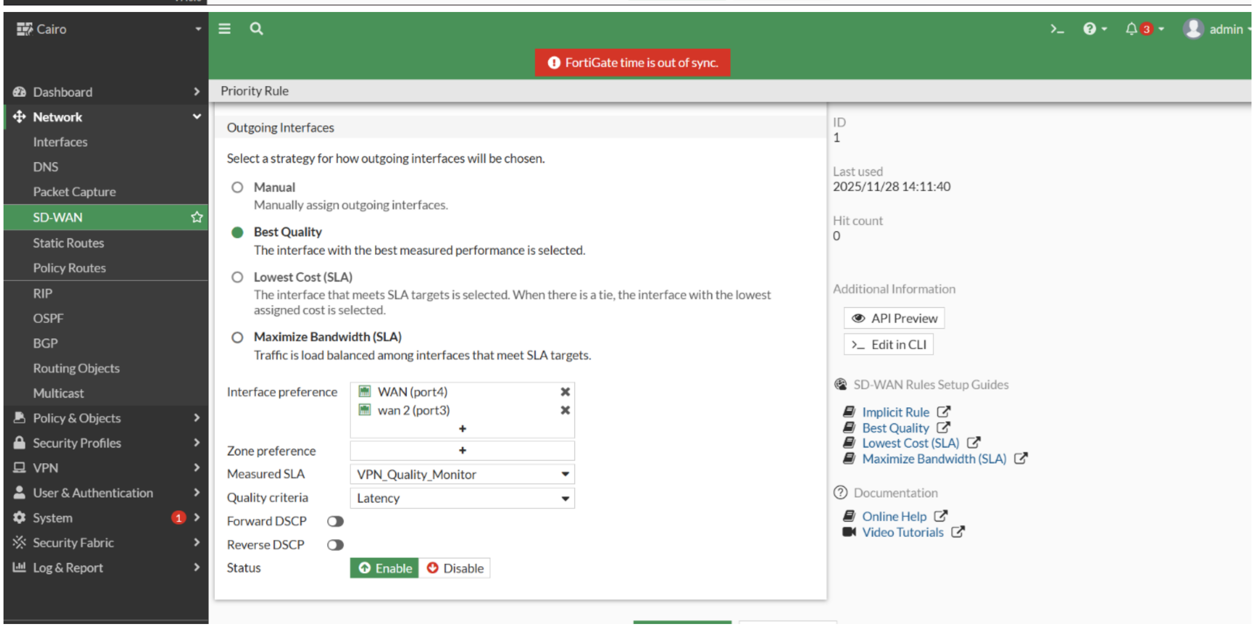
Purpose: This is the core intelligence. You define the acceptable quality targets (latency, jitter, loss) for the link carrying the VPN traffic. The FortiGate constantly checks both wan1 and wan2 against these targets.



4.Create the SD-WAN Rule (Traffic Steering Logic)

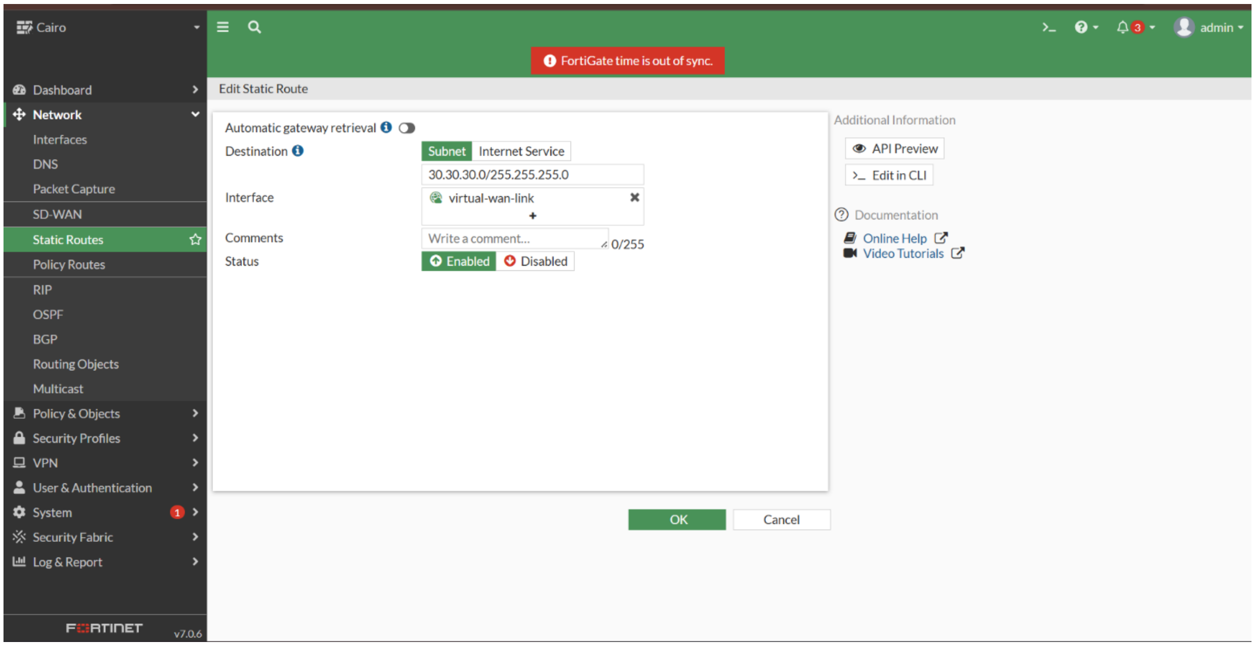
Purpose: This rule specifically directs the VPN traffic to use the **best performing** link based on the real-time data collected

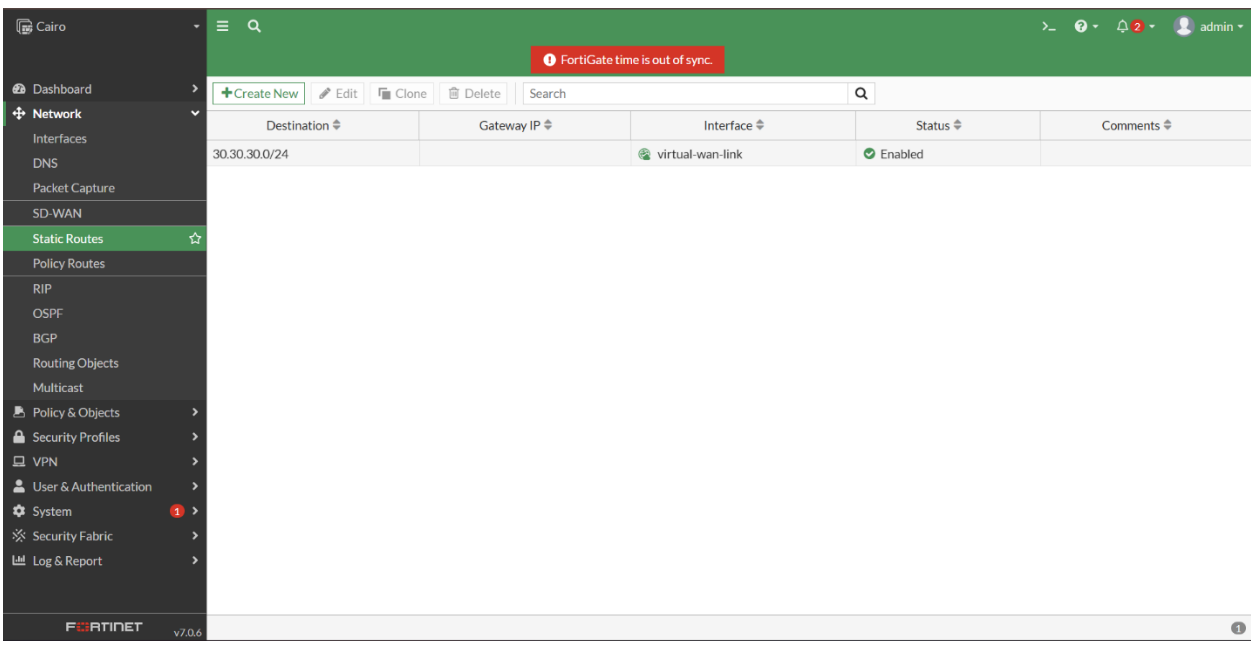



Cairo
FortiGate time is out of sync.
admin
Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report
FORTINET v7.0.6
Priority Rule
Name Optimize_Alex_VPN
Source
Source address NET 10
User group
Destination
Address NET 30
Protocol number TCP UDP ANY Specify 0
Internet Service
Application
Outgoing Interfaces
Select a strategy for how outgoing interfaces will be chosen.
Manual
Manually assign outgoing interfaces.
Best Quality
The interface with the best measured performance is selected.
Lowest Cost (SLA)
ID
1
Last used
2025/11/28 14:11:40
Hit count
0
Additional Information
API Preview
Edit in CLI
SD-WAN Rules Setup Guides
Implicit Rule
Best Quality
Lowest Cost (SLA)
Maximize Bandwidth (SLA)
Documentation
Online Help
Video Tutorials
OK
Cancel

Cairo
FortiGate time is out of sync.
admin
Dashboard
Network
Interfaces
DNS
Packet Capture
SD-WAN
Static Routes
Policy Routes
RIP
OSPF
BGP
Routing Objects
Multicast
Policy & Objects
Security Profiles
VPN
User & Authentication
System
Security Fabric
Log & Report
Priority Rule
Outgoing Interfaces
Select a strategy for how outgoing interfaces will be chosen.
Manual
Manually assign outgoing interfaces.
Best Quality
The interface with the best measured performance is selected.
Lowest Cost (SLA)
The interface that meets SLA targets is selected. When there is a tie, the interface with the lowest assigned cost is selected.
Maximize Bandwidth (SLA)
Traffic is load balanced among interfaces that meet SLA targets.
Interface preference WAN (port4)
wan 2 (port3)
Zone preference
Measured SLA VPN_Quality_Monitor
Quality criteria Latency
Forward DSCP
Reverse DSCP
Status Enable Disable
ID
1
Last used
2025/11/28 14:11:40
Hit count
0
Additional Information
API Preview
Edit in CLI
SD-WAN Rules Setup Guides
Implicit Rule
Best Quality
Lowest Cost (SLA)
Maximize Bandwidth (SLA)
Documentation
Online Help
Video Tutorials

5.Update the Static Route

Purpose: To make the SD-WAN rule effective, you must redirect the traffic bound for the remote VPN network away from a single physical interface and hand it over to the SD-WAN engine.





Conclusion

This project demonstrated the practical implementation of secure VPN solutions using FortiGate while exploring the strengths and use cases of different VPN technologies in modern network environments. Beginning with foundational concepts, the SSL VPN configuration provided secure remote access, illustrating how organizations can extend internal resources to mobile or off-site users in a controlled and authenticated manner. The IPsec VPN phase expanded on these capabilities by establishing encrypted site-to-site communication, enabling two independent networks to operate securely as a unified infrastructure. This configuration proved essential for branch connectivity, inter-office data transfer, and enterprise scalability.

The final stage integrated SD-WAN with VPN configurations to optimize performance and link utilization. By monitoring WAN health and intelligently routing VPN traffic, FortiGate's SD-WAN improved overall reliability, reduced packet loss, and ensured continuity in the presence of degraded or failing links. This demonstrated how VPN security, when combined with dynamic routing, creates a highly resilient communication environment suitable for real-world applications.

Through the implementation, testing, and performance evaluation of these solutions, the project highlighted the flexibility and robustness of Fortinet technologies. It also showed the importance of aligning VPN deployments with operational requirements—whether user access, inter-branch connectivity, or WAN performance. Overall, the project successfully achieved its objectives and provided valuable hands-on experience in designing and managing secure enterprise networks using FortiGate.